

The Sorghum Lipid Database (SoLD): population-scale lipidomics linking environmental and genetic variation in the Sorghum Association Panel

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ABSTRACT Sorghum (*Sorghum bicolor*) is a climate-resilient crop whose acclimation to nutrient limitation and low temperature likely involves extensive lipidome reconfiguration. Lipids are key membrane components, carbon and energy stores, and mediators of stress signaling, yet population-scale lipidomics data for sorghum are limited. We present the Sorghum Lipid Database (SoLD), a curated lipidomics resource from the Sorghum Association Panel grown under two field regimes, (i) a nutrient-sufficient with usual planting date environment (control) and (ii) a low-input treatment with reduced nitrogen and phosphorus, earlier planting, and no application of insecticides, herbicides, or pesticides (low-input). Using high-resolution LC–MS, we quantified 266 lipid species and found the lipidome broadly similar across trials, with four low-input-associated signatures relative to control, (i) depletion of sulfoquinovosyldiacylglycerol, (ii) triacylglycerol enrichment, (iii) phospholipid redistribution centered on phosphatidylserine, and (iv) a shift in the lysophosphatidylcholine-to-lysophosphatidylethanolamine ratio. Analyses of lipid chemical space and lipid ontology enrichment supported these compositional changes. Genomic heritability was near zero under control and 0.170 under low input, reflecting more genetic variance rather than less experimental noise. GWAS of lipid species, class sums, and class ratios recovered loci that were largely trial-specific, with candidate genes assigned by linkage disequilibrium with the peak rather than by physical distance, and prioritized both by ontology terms passing an LD-aware permutation null and by mapping onto lipid reaction branches. Control-associated loci were enriched for tyrosine catabolism and oxidative-stress functions, whereas low-input-associated loci were enriched for nutrient transport and signaling, sphingolipid metabolism, and jasmonate-related regulation, and included a chromosome 3 block carrying two LOW PHOSPHATE ROOT homologues. Thus, SoLD provides a framework connecting sorghum lipid diversity with environmental and genetic variation. All information regarding the database and the experiment is freely accessible through a Shiny application: <https://nirwan.shinyapps.io/SAP-Lipidomics-Database/>. The database enables users to move from lipid-class to individual molecular species and associated candidate loci, for hypothesis generation, comparative analyses, and prioritization of targets for functional validation.

KEYWORDS Sorghum Association Panel, Lipidomics, Genome-wide Association Studies, Triglycerides, Phospholipids, Cold stress, Nutrient stress

Introduction

Plants routinely face environmental stresses that limit yield, such as nutrient limitation and low temperature. Nutrients such as nitrogen (N) and phosphorus (P) deficiencies are important for research because they restrict overall growth, nutrient allocation, and developmental plasticity, while low temperature disrupts cell membrane fluidity, metabolism, and yield (Schlüter *et al.* 2012; Silveira *et al.* 2018; Emendack *et al.* 2021; Ding *et al.* 2019; Guo *et al.* 2025). Lipids are central to these environmental responses as cell membrane components, carbon and energy stores,

and precursors of signaling molecules. Stress-driven lipid re-modeling can alter membrane stability, photosynthetic capacity, storage-lipid abundance, and lipid-derived signaling (Bohn *et al.* 2007; Siebers *et al.* 2015; Hou *et al.* 2016; Rawat *et al.* 2021; Li *et al.* 2020a; Bhattacharya 2022; Xu *et al.* 2025). Under P limitation, plants usually replace phospholipids with non-P galactolipids and sulfolipids, including digalactosyldiacylglycerol (DGDG) and sulfoquinovosyldiacylglycerol (SQDG). Cold stress can shift acyl-chain composition, phospholipid turnover, and triacylglycerol (TG) accumulation (Essigmann *et al.* 1998; Härtel *et al.* 2000; Jouhet *et al.* 2004; Kelly and Dörmann 2002; Tan *et al.* 2018; Pfaff *et al.* 2020; Yan *et al.* 2025; Zhang *et al.* 2020). These lipid responses are followed by nutrient and stress-regulatory pathways. An example includes the SPX-domain proteins, which regulate phosphate signaling through PHR-centered networks

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(Lv *et al.* 2014). Similarly, nitrate transporters such as NRT2 and NPF mediate N acquisition and allocation, and sulfate transporters influence plastid sulfur supply for sulfolipid metabolism (Lupini *et al.* 2016; Sorgonà *et al.* 2011; Xia *et al.* 2015; Chen *et al.* 2019). In conjunction, lipid-remodeling enzymes, including non-specific phospholipases, DGATs, and SDP1-like lipases, mediate phospholipid turnover, DG production, and TG synthesis and mobilization (Bose *et al.* 2021; Ngo *et al.* 2019; Tan *et al.* 2018; Kelly *et al.* 2011; Yen *et al.* 2008). However, population-scale lipidomics resources linking lipid remodeling, nutrient-stress, and natural genetic variation remain scarce in crops, particularly in sorghum.

Sorghum is a stress-resilient cereal crop with large genetic diversity. The Sorghum Association Panel (SAP) is a collection of sorghum lines with substantial genomic variation that captures broad botanical, geographic, and genetic diversity (Casa *et al.* 2008; Boatwright *et al.* 2022). This makes it ideal for studying natural variation due to environmental stresses. For lipidomics, however, population-scale interpretation requires careful data processing and annotation. LC-MS lipidomics can be affected by run-order drift, instrumental batch effects, and field-layout artifacts, while lipid annotation and naming can vary across studies. Here, we have used a QC-based normalization method, SERRF; spatial correction methods, SpATS; and a spectral annotation database, GNPS, that improve reproducibility and biological interpretation (Fan *et al.* 2019; Rodriguez-Alvarez *et al.* 2018; Wang *et al.* 2016; Molenaar *et al.* 2019).

A second challenge in population-scale lipidomics linked to GWAS is interpreting large candidate-gene sets. GWAS across many lipid species, class sums, and class ratios can yield hundreds to thousands of nearby genes, but LD-assigned genes are not independent. A single association can tag multiple neighboring genes, including tandem arrays with similar annotations, so standard gene-level enrichment can misinterpret one locus as pathway-level evidence. This issue led to GWAS-specific gene-set and interval-enrichment tools such as INRICH, SNP2GO, and Gowinda, which adjust for LD, candidate-SNP structure, or gene-level biases (Lee *et al.* 2012; Szkiba *et al.* 2014; Kofler and Schlötterer 2012). Here, we extend this approach by testing Gene Ontology (GO) enrichment against an LD-aware interval-permutation null and by distinguishing terms supported by multiple independent genomic intervals from those driven by single annotated loci.

Here, we present the Sorghum Lipid Database (SoLD), a curated lipidomics and GWAS resource for the SAP. Leaf samples collected between the V6 and V8 stages were profiled under two field regimes: a control environment (CTL), grown in 2019 with sufficient nutrients and standard planting, and a low-input regime (LIN), grown in 2022 with reduced N and P, earlier planting to impose early-season cold-associated stress, and minimal chemical management, including no insecticide, herbicide, or pesticide application. Since CTL and LIN were grown in different field years, we interpret their contrasts conservatively as between-trial differences rather than direct causal treatment effects.

We combine LC-MS lipidomics, compositional analysis, lipid ontology, GWAS of individual species and class-level traits, LD-aware GO enrichment, LINEX-based reaction mapping, and an interactive Shiny application. This framework identifies four major LIN-associated lipid signatures: SQDG depletion, TG enrichment, PS-centered phospholipid redistribution, and altered LPC/LPE balance. It also prioritizes candidate func-

tions linked to nutrient acquisition, DG-TG cycling, phospholipid turnover, sphingolipid metabolism, and stress-associated membrane remodeling, including representative loci such as DGAT1 (SORBI_3010G170000) for TG accumulation, a SULTR3-family sulfate transporter (SORBI_3006G244000) for SQDG-related sulfur supply, and an SPX-domain phosphate-status (SORBI_3001G028500) candidate. Together, SoLD provides a resource for exploring sorghum lipid variation, candidate loci, and accessions for future validation. The interactive SoLD Shiny application enables exploration of all datasets and results and is available at <https://nirwan.shinyapps.io/SAP-Lipidomics-Database/>.

Results

QC and normalization yielded comparable LC-MS performance across conditions

We profiled early-stage lipids from SAP grown in two field conditions defined earlier. The workflow (Supplementary S1 Fig) comprised tissue collection and extraction, high-resolution LC-MS lipidomic profiling, GNPS-assisted lipid annotation, post-acquisition data correction with SERRF and SpATS (Fan *et al.* 2019; Rodriguez-Alvarez *et al.* 2018), and data dissemination via the SoLD Shiny app (<https://nirwan.shinyapps.io/SAP-Lipidomics-Database/>).

Run-order quality control showed stable analytical performance (Supplementary S2 FigA–B), with the total ion current (TIC) and the summed signal intensity per sample meeting expected values across system checks, internal standards, and blanks. TIC variability among QC samples was low (7.5% CV for CTL and 6.0% CV for LIN). SERRF normalization improved technical precision, reducing median QC-RSD from 3.6% to 2.0% in CTL and from 1.6% to 0.3% in LIN. The proportion of features with QC-RSD below 30% rose from 75.7% to 94.1% and from 77.3% to 89.3% respectively (Supplementary S2 FigC–D).

PCA of normalized data showed tighter sample clustering than unprocessed data (Supplementary S2 FigE–F). After spatial correction, the leftover variation in MGDG(18:3/18:3) is scattered randomly across the field rather than forming rows, columns, or patches, indicating that position in the field no longer drives the signal (Supplementary S2 FigG–H). Thus, the post-correction, lipid signal distributions remained similar across cohorts, and the adjusted trait matrix was used for all subsequent analyses.

Genotypic lipid variation is largely independent of botanical race, and heritable variation is confined to LIN

SAP is structured by botanical race and geographic origin (Casa *et al.* 2008; Boatwright *et al.* 2022). Next, we tested whether lipid variation is affected by race or marker-based genetic clusters ($K = 6$). Using Kruskal-Wallis with BH correction within trials (Supplementary Table S1), we found little association. No lipid-class sum was significant for race in either trial. The genetic cluster was significant only for PS under LIN ($q = 1.16 \times 10^{-6}$, $\epsilon^2 = 0.104$; Fig. 1). PS is carried by a single species under LIN, PS(18:0/18:2), so the class sum and that species are the same measurement. The same class-level test under CTL, where PS is carried by two species, was not significant ($\epsilon^2 = 0.020$, $q = 0.358$). Furthermore, none of the species-level tests were significant under CTL, and nine were significant under LIN, eight of them by genetic cluster alone and one, 11,14,17-eicosatrienoic acid, by both genetic cluster and botanical race (Table 2; Supplementary Table S2 (per-species structure and h^2)). PCA of class composition likewise showed minimal structure, with groupings

explaining $\leq 1.6\%$ of variation (Supplementary S3 Fig). Thus, ancestry is not a confounder.

Subsequently, we quantified genetic variance in the lipidome by estimating genomic heritability (h^2) for every species detected in each trial (214 under CTL, 216 under LIN) and compared the trials on the 163 species detected in both and measured across the same 351 genotypes (Supplementary Table S2, per-species paired). Under CTL, the median h^2 was near zero, and only 11 of those species had heritability that we could distinguish from zero. Under LIN, the median was 0.170, and 53 species did. The same direction held for species by species comparison, with h^2 higher under LIN for 131 species, higher under CTL for 21, and equal for 11 (Supplementary S4 Fig), and for class by class comparison where every class was higher under LIN except TG and PA (Supplementary Table S2, structure-conditioned). Two checks indicate that this is not an artifact of ancestry or of which genotypes were sampled. Fitting the three leading eigenvectors of the relationship matrix as fixed effects, which removes variation between genetic clusters, left 51 of the 66 LIN-heritable species still heritable (Supplementary Table S2, structure-conditioned). Re-estimating after dropping each botanical race in turn and after twenty random 80% subsamples of the panel gave 14 to 26 heritable species under CTL and 43 to 87 under LIN on the full per-trial sets. The two ranges never overlap, so LIN exceeded CTL in every run (Supplementary Table S2, ancestry robustness).

Moreover, heritability increases unevenly across classes (Table 1). Under CTL, it is largely confined to TGs, which account for 19 of the 67 TG species and for 19 of the 20 heritable species in that trial. Under LIN, it concentrates in the plastid membrane lipids, with 4/5 MGDG, 6/10 SQDG and 4/7 DGDG, the highest proportions of any class with more than one species. MGDG and SQDG also carry the largest absolute shifts in %TIC of any class apart from TG, falling by 2.6 and 1.5 percentage points while TG rises by 2.3, indicating that both compositional and genetic changes are centered on the photosynthetic membrane lipidome. TG is the exception, being heritable in both trials (19/67 under CTL, 25/66 under LIN). PS, PA, and LPE each include a single species, so they reflect that species rather than the class.

The near-absence of heritable species under CTL reflects low genetic variance rather than high measurement noise. Median residual variance was almost identical between trials (0.045 under CTL, 0.044 under LIN), and total variance was essentially unchanged, while median genetic variance rose from effectively zero to 9.0×10^{-4} and was higher under LIN for 25 of 26 class sums (Supplementary Table S2, class sums). Aggregating species into class sums did not recover heritability in either trial. In each trial, only two of the 13 class sums had intervals excluding zero, and in each trial, one of those two classes contained a single species, so the class-level test is less powerful than the species-level one rather than more. Genotype, therefore, accounts for far more of the leaf lipidome under LIN than under CTL, which is the asymmetry that the GWAS results below follow.

Lipidome overview for CTL and LIN SAP

We quantified a total of 266 lipid species, with 164 detected in both trials, 50 only in CTL, and 52 only in LIN, where “trial-specific” indicates detection above the threshold in one trial rather than absence in the other (Supplementary S5 Fig). Per-species values for all species are browsable in the SoLD Shiny application. TGs and PCs were the most diverse classes in both environments (TG: 67 in CTL, 66 in LIN; PC: 36 in CTL, 38 in

Table 1 Heritable lipid species are concentrated under LIN and fall mainly on the plastid membrane classes. Genomic heritability was estimated for every lipid species detected in a trial, on one genomic relationship matrix. Species, the number of species of that class in each trial; median h^2 , the class median; $h^2 > 0$, the number whose 95% profile-likelihood interval excludes zero; conditioned, the number still excluding zero under LIN once the three leading eigenvectors of the relationship matrix are fitted as fixed effects, which removes variance between genetic clusters. The two counts are not nested, because removing structure can sharpen a borderline signal as well as remove one. Classes follow the frozen species set (Supplementary Table S3), so the 13 focal classes are named and everything else is Other. The paired CTL–LIN comparison in the text uses only the 163 species measured in both trials; this table describes each trial as measured. [†]Represented by a single species, whose entry describes that species rather than a class. [§]Fatty acids, ether lipids, sphingolipids and other minor categories.

Class	Species		CTL		LIN		
	CTL	LIN	median h^2	$h^2 > 0$	median h^2	$h^2 > 0$	conditioned
TG	67	66	0.084	19	0.190	25	28
PC	36	38	0.000	0	0.095	7	7
Other [§]	39	45	0.000	0	0.128	13	13
PE	14	15	0.000	0	0.174	4	2
DG	16	15	0.000	0	0.066	0	2
SQDG	11	10	0.000	0	0.341	6	5
DGDG	7	7	0.000	0	0.202	4	2
MGDG	5	5	0.000	0	0.456	4	4
MG	8	5	0.000	0	0.098	0	0
LPC	4	3	0.000	0	0.164	1	1
PG	3	3	0.000	0	0.158	1	0
PA [†]	1	2	0.554	1	0.000	0	0
PS [†]	2	1	0.000	0	0.382	1	1
LPE [†]	1	1	0.000	0	0.020	0	0
Total	214	216		20		66	65

LIN; Supplementary S5 Fig). The species contributing most to between-genotype variance in each trial are listed in Supplementary Table S4 (top-variance lipids).

Furthermore, we summarized lipid abundance as %TIC (mean share of total annotated signal per sample), so values represent the relative signal in each condition. In both trials, the glycerolipids and glycerophospholipids dominated the annotated signal (CTL: 65.4% and 30.3%; LIN: 64.3% and 30.7%), while the prenol lipids were the only other LIPID MAPS category above 1% (3.4% and 4.2%) (Fig. 2A). The sterol lipids show the largest proportional change of any category, rising 5.1-fold from 0.04% to 0.22% of the annotated signal, almost all of it β -sitosterol (0.039% to 0.178%). Both values are small, so the change cannot substantially alter the bulk composition.

Within the galactolipids and phospholipids, MGDG contributed the most with 32.7% (CTL) and 30.1% (LIN), while PC contributed 24.2% and 23.3%. DGDG (12.1%, 12.2%) and DG (13.0%, 13.8%) followed. TG increased from 2.6% (CTL) to 4.9%

Table 2 The individual lipid species associated with ancestry, all of them under LIN. Every species detected in a trial was tested against botanical race and against marker-based genetic cluster (Kruskal–Wallis, Benjamini–Hochberg corrected within trial and grouping; 214 species under CTL, 216 under LIN). No species reached $q < 0.05$ under CTL for either grouping, and the five below are the complete set under LIN. Genomic heritability for the same species is given beside the tests, before and after the three leading eigenvectors of the genomic relationship matrix are fitted as fixed effects; conditioned, whether the 95% profile-likelihood interval still excludes zero. [‡]This feature carries most of its signal in a small number of accessions and lies near the detection limit, so its association is reported for completeness and is not interpreted. Full per-species results are in Supplementary Table S2 (per-species structure and h^2).

Species	Class	Genetic cluster		Botanical race		Heritability		
		ϵ^2	q	ϵ^2	q	h^2	h^2 cond.	cond. > 0
β -Sitosterol	–	0.095	7.7×10^{-5}	0.011	0.625	0.676	0.644	yes
PS(18:0/18:2)	PS	0.088	1.2×10^{-4}	0.018	0.625	0.382	0.294	yes
11,14,17-Eicosatrienoic acid,(Z,Z,Z)-	–	0.074	6.5×10^{-4}	0.113	0.007	0.816	0.782	yes
Fucosterol	–	0.073	6.5×10^{-4}	0.013	0.625	0.404	0.322	yes
Cycloartenol acetate	–	0.060	4.0×10^{-3}	0.031	0.597	0.688	0.662	yes
PC(18:0/20:2)	PC	0.048	0.021	0.000	0.643	0.102	0.028	no
PE(18:0/18:3)	PE	0.047	0.021	0.000	0.671	0.358	0.350	yes
AEG(o-15:2/15:0)	AEG	0.046	0.023	0.043	0.433	0.392	0.290	no
SQDG(16:0/16:0)	SQDG	0.041	0.043	0.049	0.428	0.486	0.454	yes

(LIN), while SQDG decreased from 2.8% to 1.3%. Overall, these lipids comprised 94.7% (CTL) and 93.5% (LIN) of annotated signals, with the remainder from sterols, terpenoids, prenols, fatty acids, and other minor categories (Fig. 2B; Supplementary Table S4, composition %TIC). Class-level contrasts between trials on the centered and additive log-ratio scales are given in Supplementary Table S4 (CLR and ALR contrasts). These class shares do not depend on which accessions are in the panel. Deleting each botanical race group in turn and repeatedly deleting 20% of accessions at random moved the mean %TIC of the largest class, PC, by at most 0.10 percentage points under CTL and 0.18 under LIN, with every other class moving less (Supplementary Table S3, composition stability).

Additionally, two minor pools show large changes masked in the bulk profile. PS increases from 0.014% to 0.045% of annotated signal, and its correlation with PC flips between trials ($r = 0.43$ in CTL vs. $r = -0.34$ in LIN; Supplementary Table S4, CLR correlation delta; Supplementary S6 Fig). Among lysophospholipids the two classes move in opposite directions. LPE rises from 0.006% to 0.014% of annotated signal while LPC falls on both log-ratio scales (CLR -0.14 , ALR -0.62 ; Supplementary Table S4, CLR and ALR contrasts), so the LPC/LPE ratio halves from 23.2 to 11.1. Both pools stay below 0.2% in both trials.

After using PCA to assess how genotype and botanical race structure the diversity panel, we performed a similar analysis on the lipidomic data. Within each trial, PCA of major glycerolipid and glycerophospholipid pools showed that lipid class was the main driver of species-level variation (Supplementary S7 Fig). Results for the remaining lipid classes are available in the Shiny app. TG species formed a distinct cluster separated from most membrane lipids in both CTL and LIN, though along different principal axes. Galactolipids and glycerophospholipids partly overlapped but occupied class-enriched regions, supporting class-level aggregation for downstream compositional and reaction-network analyses.

We next asked whether the observed class shifts were accompanied by shared structural and biophysical properties using

LION, a lipid ontology that groups species into such categories (Molenaar *et al.* 2019). LION ranking analysis across the 164 shared lipid species returned ten significant terms, five higher and five lower with respect to LIN ($q < 0.05$; Fig. 2C; Supplementary Table S5, LION enrichment). The three strongest terms, *triacylglycerols*, *lipid storage*, and *lipid droplet*, are annotated on the same 44 species and are therefore reported as one result in three ways. The remaining terms higher in LIN were *headgroup with neutral charge* (58 species) and *glycerolipids* (61 species). A caution is that LION's glycerolipid term covers only TG, DG, and MG because the galactolipids are absent from the ontology, and that pool rises from 16.9% to 19.7% of annotated signal even though the broader glycerolipid in Fig. 2A does not. Lower in LIN were *membrane component* (65 species), *low lateral diffusion* (11 species), *lipid-mediated signalling* (23 species), *PC(38:2)* (3 species), and *C20:1* (4 species). The last three rely on few annotated lipids, and we do not interpret them individually. LION is built on LIPID MAPS and carries no terms for the plant galactolipids or sulfolipid, placing the classes with the strongest compositional signal outside the enrichment universe. A caution is that the difference in year, field, and acquisition batch applies to every term shown.

Finally, to describe class chemistry beyond abundance, we summarized each lipid class in a reduced chemical space using the abundance-weighted mean total carbon number and double-bond count across molecular species (Supplementary S8 Fig; Supplementary Table S5, chemical space). Carbon number cleanly separates the lipid. Lysophospholipids and monoacylglycerols have 16–18 carbons, diacyl lipids have 33–37, and TG has 51.0 (CTL) and 54.6 (LIN). Diacyl unsaturation varies independently of chain length, increasing from PG (0.6 double bonds) to PC (2.1) to MGDG (5.7). Across trials, PE shows the largest unsaturation increase (2.7 to 3.4 double bonds) among classes with >10 species. Otherwise, aside from a higher TG carbon number, all classes except PS change by <1 carbon and <0.25 double bonds. PS is not comparable (two species to one). Thus, within-class chemistry is largely stable, and CTL–LIN

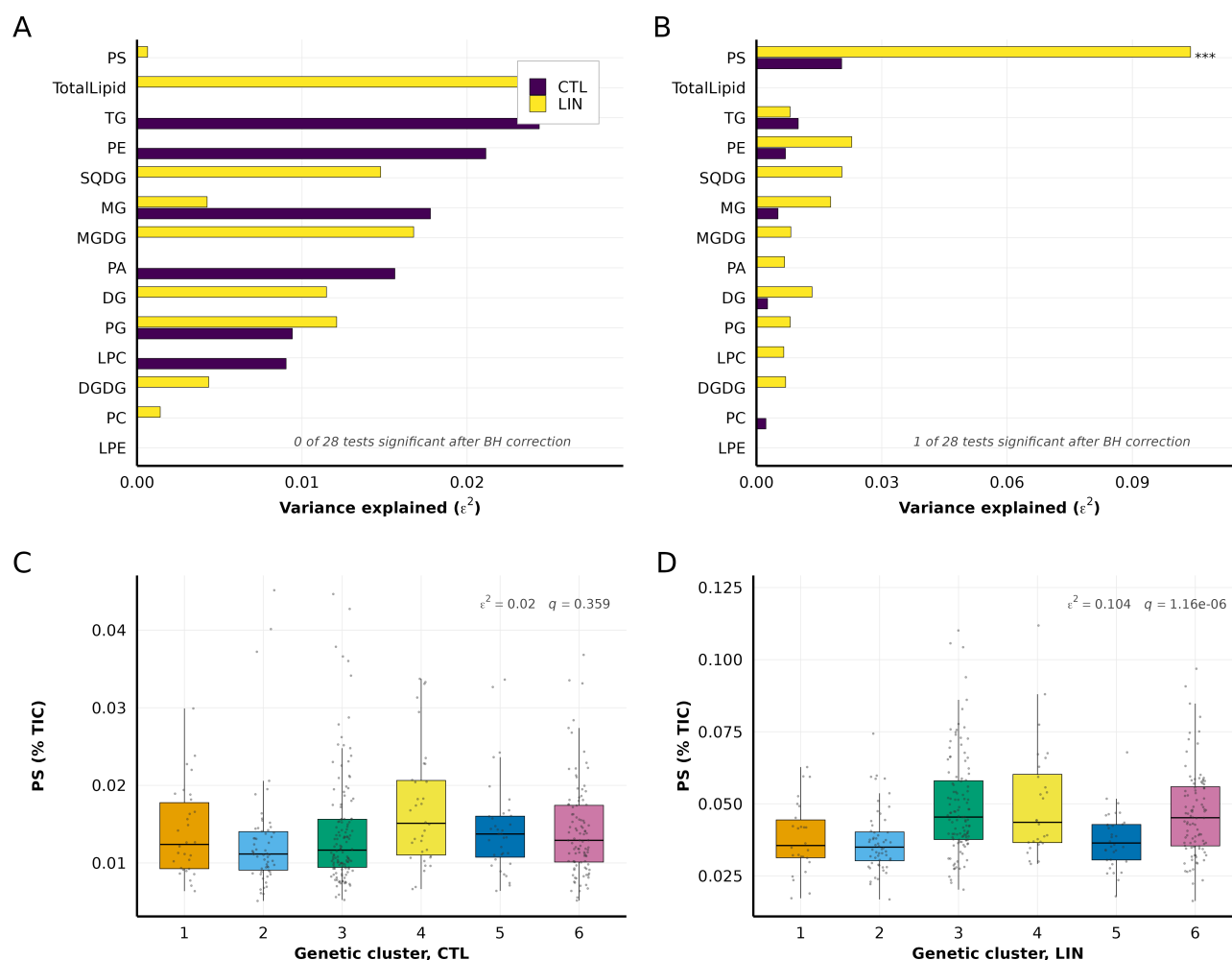


Figure 1 Lipid-class variation is largely independent of botanical race and of population structure, except for PS. (A,B) Variance explained (ϵ^2) by botanical race (six groups) and by marker-based genetic cluster ($K = 6$), for total lipid signal and the 13 lipid classes, computed within each field trial. Stars mark tests significant after Benjamini–Hochberg correction. **(C,D)** PS abundance (% TIC) by genetic cluster in each trial. Each panel keeps its own scale, since the comparison is between clusters within a trial. PS under LIN is the only trait of the 28 tested that is associated with cluster membership.

differences mainly reflect shifts in class proportions, not chain composition.

Compared with CTL, four population-wide shifts characterize the LIN lipidome: (i) SQDG depletion, (ii) TG enrichment, (iii) a phospholipid rebalancing centered on PS, and (iv) an LPC/LPE balance shifted toward LPE. All four rest on the abundance and composition analyses. LION corroborates only the TG shift, its remaining terms tracking the same glycerolipid-versus-membrane contrast. The chemical-space summary adds no further shift, because within-class chain composition is largely stable and the CTL–LIN difference sits in class proportions rather than chain length or unsaturation. Because the trials also differ by year, field, planting date, and analytical batch, these shifts characterize LIN as a whole and cannot be attributed to just nutrient supply and cold.

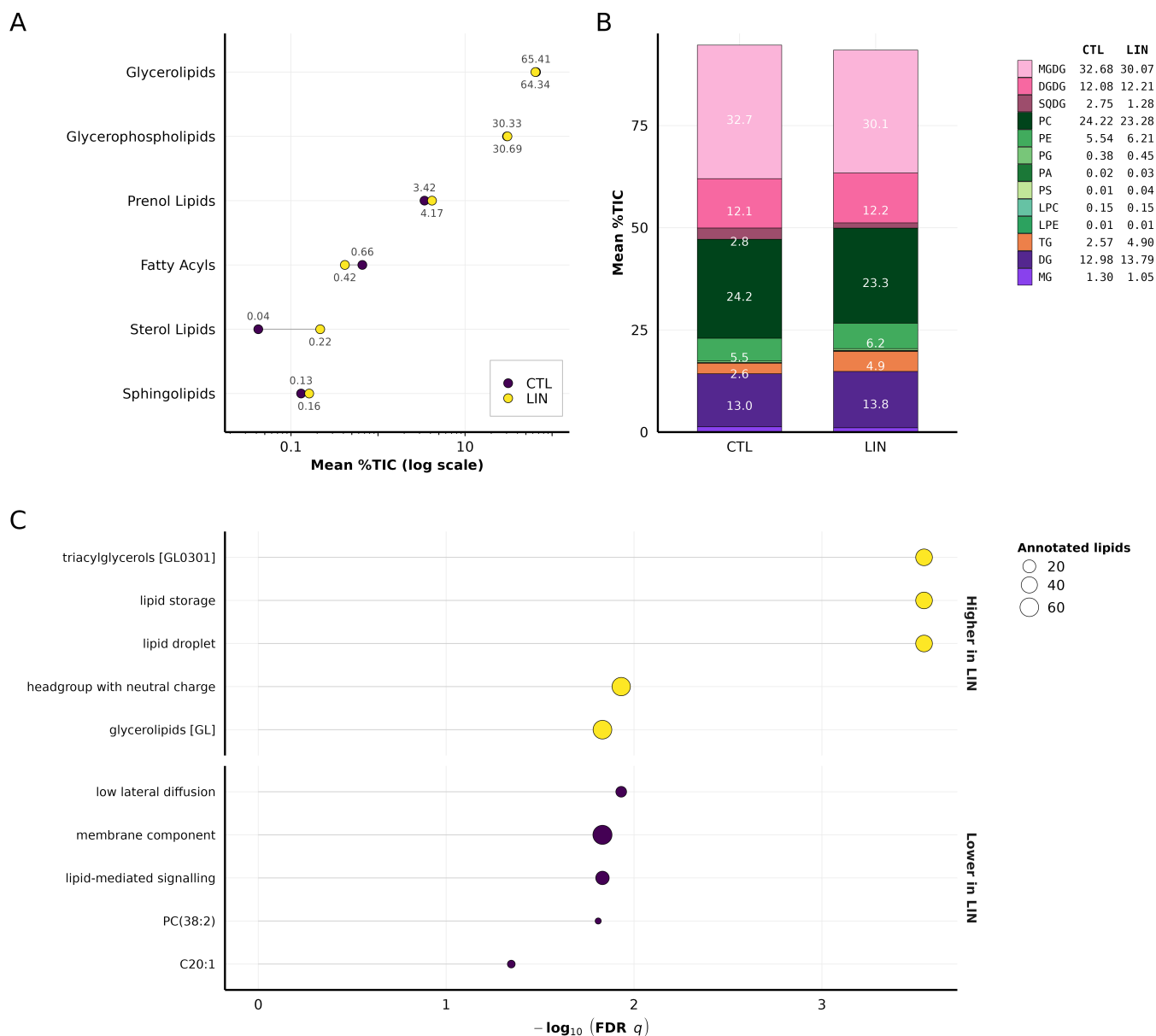
The reaction-network and GWAS sections that follow ask which reactions and which loci accompany them. Candidate genes follow for the TG and SQDG shifts. The PS and lysophospholipid changes sit on pools under 0.2% of TIC and are not

resolved to candidates, although PS is the one class whose abundance tracks genetic cluster (Fig. 1C,D).

GWAS recovers more genomic regions under LIN and more trait-sharing per region under CTL

We ran separate GWAS for each individual lipid trait, lipid-class sum, and lipid-class ratio in both trials (1,102 in total: 225 individual species and 324 class sums and ratios under CTL, 229 and 324 under LIN), using a univariate linear mixed model with the first three genotype PCs to account for population structure (Fig. 3). SNPs surpassing the Bonferroni genome-wide significance threshold ($p \leq 8.19 \times 10^{-9}$ for CTL and 8.22×10^{-9} for LIN) were mapped to candidate genes using the LD-based cutoff of 0.4. Under CTL, this identified 1,100 candidate genes for individual lipids and 115 for class sums and ratios (Supplementary Table S6). Under LIN, the same procedure identified 4,319 and 812 candidate genes (Supplementary Table S6). Manhattan and QQ plots for every lipid phenotype in both trials are available in the SoLD Shiny application.

Candidate gene counts do not equal signal counts because



one significant SNP tags all genes in its LD window. After binning candidates into 250 kb windows, there were 378 distinct windows under CTL and 970 under LIN. So, LIN captures $\sim 2.6\times$ more loci, not the nearly four-fold suggested by gene counts. Hence, CTL loci average 2.9 candidate genes, while LIN loci average 4.5. Candidates also differ in phenotype recurrence. Under CTL, 612 of 1,100 individual-lipid candidates were linked to more than one phenotype (median 2, maximum 25), against 1,302 of 4,319 under LIN (median 1, maximum 12). The class-sum/ratio layer shows the same pattern (CTL median 3, maximum 64; LIN median 1, maximum 20). Thus, CTL

yields fewer loci with many lipid associations, while LIN yields more loci with few. Because lipid phenotypes are inherently highly correlated by design, these recurrence counts serve as descriptive summaries rather than formal tests against a null hypothesis; the cross-trial comparison presented in the next section is evaluated explicitly.

In the CTL individual-lipid layer, the strongest association was *SORBI_3001G522600* (CW-type domain-containing protein; $p = 4.50 \times 10^{-35}$; $r_{\max}^2 = 1.0$), and the most recurrent locus was *SORBI_3006G040500* (phospholipid-transporting ATPase; 25 phenotypes; $p = 1.01 \times 10^{-26}$). In the CTL sum/ratio

layer, the strongest association was SORBI_3010G136200 (SLH domain-containing protein; $p = 4.28 \times 10^{-23}$; $r^2_{\max} = 0.91$), and the most recurrent locus was SORBI_3003G318650 (64 phenotypes; $p = 3.37 \times 10^{-12}$; $r^2_{\max} = 0.90$) (Table 3). In the LIN individual-lipid layer, the strongest association was observed for SORBI_3004G130500, an unannotated candidate gene ($p = 2.44 \times 10^{-73}$; $r^2_{\max} = 1.0$). The most recurrent individual-lipid locus was SORBI_3010G064200 (1,3-beta-glucan synthase; 12 phenotypes; $p = 3.40 \times 10^{-16}$; $r^2_{\max} = 1.0$). In the LIN sum/ratio layer, the strongest association and the most recurrent locus were both SORBI_3004G004800 (soluble acid invertase; 20 phenotypes; $p = 2.61 \times 10^{-21}$; $r^2_{\max} = 0.78$) (Table 3).

Table 3 lists the five most recurrent loci in each trial and trait layer, counted per signal rather than per gene, with the number of genes each signal tags alongside it. The full candidate-gene lists are browsable in the SoLD Shiny application. The contrast between the trials is not only a matter of how the candidate tables are counted. Genomic heritability, as estimated above, is higher under LIN, and that estimate involves no LD windows and no gene counting. More heritable traits are also easier to map, so the larger LIN candidate list follows from the heritability result rather than being independent evidence for it.

LD-aware GO enrichment reduces the candidate lists to a small number of interpretable terms

Both candidate lists were too large to inspect gene by gene, so we tested Gene Ontology (GO) biological-process and molecular-function terms for over-representation within each lipid class and trials, using a permutation null that preserves LD-block structure. This distinguishes terms supported by multiple independent intervals from those driven by a single LD block.

In total, 101 of 103 terms passed the LD-aware null ($q_{LD} < 0.05$). Because GO terms are nested and class sums and ratios share candidate genes, we merged terms with identical gene sets, leaving 50 term-gene-set combinations across 38 distinct terms (Supplementary Table S7, loci collapsed). Of these, 21 draw on ≤ 30 background genes and are shown in Fig. 4; the rest were too broad to interpret (*zinc ion binding*, 1,015 genes; *transmembrane transport*, 526) and pointed to no lipid-specific process. Fifteen of the 21 are supported by two or more independent intervals, so most are not the product of a single LD block.

Those 21 terms are unevenly split between trials, with 19 under LIN and 2 under CTL, the same asymmetry that the candidate counts show. The two CTL terms are both related to tyrosine catabolism, derived from TG traits. The LIN terms center on nutrient transport and signaling, with cellular response to phosphate starvation and nitrate assimilation from SQDG and fatty-acid traits, along with sphingolipid metabolism and jasmonate-related defense regulation from AEG. They are carried mostly by SQDG and PC traits (8 and 5 of the 19), with fatty acids and TG contributing 3 each. Enrichment therefore narrows five thousand candidate genes to a short list which could be used for functional validation.

Only class-level traits share loci between trials more often than chance

The two trials recovered largely different candidate lists, but different GO terms cannot indicate whether the underlying genetics differ. We therefore counted how many candidate genes the two conditions share and compared that count with the chance expectation from the 34,027 *Sorghum bicolor* v3.1 gene models, both at gene resolution and after collapsing linked genes into 100, 250,

and 500 kb windows (Fig. S9 Fig). Overlap was tested with a hypergeometric test and summarized as fold enrichment over chance and as the Jaccard index, the proportion of all candidate genes appearing in both conditions.

At the gene level, the two candidate sets overlapped far more than chance, with 354 genes shared against 148 expected ($2.39\times$, $p = 4.7 \times 10^{-59}$), and each trait layer showed the excess on its own ($2.07\times$ for individual lipids, $6.56\times$ for class sums and ratios; Table 4; Supplementary Table S8). The shared genes were nonetheless a small fraction of the total, with the Jaccard index between the CTL and LIN sets, ranging from 0.020 to 0.069. Within the individual-lipid layer, the excess is carried by TG and PC ($1.63\times$ and $3.54\times$, $q < 0.01$), while fatty acids overlap no more than chance, and the remaining four classes tested share no genes at all (Supplementary Table S8, by lipid class). This mainly reflects LD because each association maps every gene in its window, and gene-dense regions are mapped repeatedly in both trials. Because no window size is objectively correct for collapsing linkage, we repeated the test at 100, 250, and 500 kb and read the trend across them as the result. For individual lipids, the excess fell at every step, from $2.07\times$ at gene level to $1.15\times$ at 100 kb and $1.04\times$ at 250 kb, where it was no longer significant ($p = 0.23$), so loci recovered in the two trials are no more concordant than two independent samples of the same genome. The class-sum/ratio layer fell more slowly and was still enriched at 250 kb ($2.36\times$, $p = 0.037$), so class-level traits reproduce across trials where individual species do not. Broken down by lipid class, the overlap is concentrated in TG and PC ($1.63\times$ and $3.54\times$, $q = 0.009$). Fatty acids overlap no more than chance and four classes share no genes at all, including SQDG despite its 396 LIN candidates. (Table 5). Recovering the same gene rarely meant recovering it for the same lipid, at 90 of 354 shared genes (25.4

What recurrence exists comes from very small regions. The 18 genes shared in the class-sum/ratio layer fall into two contiguous LD blocks, a ten-gene block on chromosome 10 (SORBI_3010G150325–SORBI_3010G151325) that contains the CMP-sialic acid transporter SORBI_3010G151000 discussed below, and an eight-gene block on chromosome 1 (SORBI_3001G168800–SORBI_3001G171600). The CTL and LIN lipid GWAS therefore recover largely non-overlapping genetic architectures. For individual lipid species, the two trials share no more loci than chance, and for class sums and ratios, reproducibility depends on two regions rather than on a dispersed set of loci. Candidate gene counts are inflated several-fold by LD in both layers.

LINEX-based reaction mapping, reaction-balance scoring, and GWAS support

To place lipid associations in a biochemical context, we used LINEX2 (Rose *et al.* 2023), which integrates curated lipid-reaction data with measured lipid abundances to identify biochemical connections among lipids and enzymes (Morgat *et al.* 2016). Since LINEX2 does not provide a sorghum reference, we used rice instead. Therefore, the resulting network is considered a set of testable hypotheses. The resulting network (Fig. 5A) revealed four candidate branches linking DG/MG, TG, and lysophospholipid pools: (i) PC+DG $\leftrightarrow$ LPC+TG (LCAT*), (ii) TG $\leftrightarrow$ DG (PNPLA1-labeled lipase branch), (iii) PE+DG $\leftrightarrow$ TG+LPE (LRO1/PDAT-like transfer), and (iv) DG+MG $\leftrightarrow$ TG (PNPLA3-labeled branch). LINEX2 returned an enzyme assignment for three of these but not for

Table 3 The most recurrent GWAS loci under CTL and LIN. The five loci associated with the largest number of phenotypes in each trial and trait layer. A locus is one signal, defined as all candidate genes sharing a trial, trait layer, chromosome and best p -value, so the genes tagged by one lead SNP occupy a single row. Interval, the span of the tagged genes in Mb; Genes, the number of candidate genes the signal tags; Gene, the first annotated gene in that interval, given as a label for the locus rather than as a causal assignment; Traits, the number of lipid phenotypes associated with the locus. Recurrence counts are descriptive summaries of the candidate tables and not a test against a null expectation, since lipid phenotypes are correlated by construction.

Trait layer	Interval (Mb)	Genes	Gene	Putative function	Traits	P
CTL						
Individual lipids	6:25.52–25.57	1	SORBI_3006G040500	Phospholipid-transporting ATPase	25	1.01×10^{-26}
	5:61.66–61.75	2	SORBI_3005G148500	FBD domain-containing protein	13	1.55×10^{-16}
	10:32.68–32.68	1	SORBI_3010G140800	Uncharacterized protein	13	1.24×10^{-14}
	9:20.91–20.93	1	SORBI_3009G092100	Uncharacterized protein	12	1.34×10^{-17}
	5:24.10–24.13	2	SORBI_3005G110120	Uncharacterized protein	12	8.19×10^{-15}
Sums / ratios	3:64.63–64.63	1	SORBI_3003G318650	Uncharacterized protein	64	3.37×10^{-12}
	2:73.01–73.28	6	SORBI_3002G371500	RRM domain-containing protein	56	1.59×10^{-11}
	9:50.91–50.98	3	SORBI_3009G152800	MFS transporter domain protein	53	1.95×10^{-10}
	10:20.53–20.94	7	SORBI_3010G136200	SLH domain-containing protein	34	4.28×10^{-23}
	10:39.42–39.42	1	SORBI_3010G144500	Uncharacterized protein	34	1.31×10^{-21}
LIN						
Individual lipids	3:29.13–29.18	1	SORBI_3003G169600	Mitochondrial carrier protein	12	6.98×10^{-67}
	10:5.08–5.09	1	SORBI_3010G064200	1,3-beta-glucan synthase	12	3.40×10^{-16}
	7:28.03–28.04	1	SORBI_3007G102400	Aminotransferase-like domain protein	8	1.97×10^{-12}
	10:43.19–43.39	9	SORBI_3010G150325	F-box domain-containing protein	7	1.41×10^{-18}
	2:14.26–14.49	13	SORBI_3002G115900	Uncharacterized protein	7	3.76×10^{-13}
Sums / ratios	4:0.42–0.78	27	SORBI_3004G004800	Soluble acid invertase	20	2.61×10^{-21}
	3:7.54–7.78	15	SORBI_3003G087100	Protein MAO HUIZ1 4, chloroplastic	19	4.46×10^{-20}
	10:43.37–43.77	10	SORBI_3010G150325	F-box domain-containing protein	16	3.73×10^{-11}
	1:7.56–8.03	33	SORBI_3001G101700	SS18 N-terminal domain protein	15	7.42×10^{-14}
	1:8.02–8.03	2	SORBI_3001G104800	Uncharacterized protein	15	1.38×10^{-13}

Table 4 Cross-trial overlap of GWAS candidates at gene and locus resolution. Overlap between CTL and LIN candidate sets, tested against a background of 34,027 annotated gene models (gene level) or of all genome windows containing at least one gene (locus level). Fold, observed/expected shared; J , Jaccard index; P , hypergeometric (right-tail) test. Bold rows are the resolutions at which LD blocks are effectively collapsed.

Resolution	Trait layer	CTL	LIN	Shared	Exp.	Fold	J	P
Gene	All layers	1165	4323	354	148.0	2.39	0.069	4.7×10^{-59}
Gene	Individual lipids	1100	4319	289	139.6	2.07	0.056	3.6×10^{-35}
Gene	Class sums / ratios	115	812	18	2.7	6.56	0.020	2.9×10^{-10}
100 kb	Individual lipids	521	1562	193	168.0	1.15	0.102	8.0×10^{-3}
100 kb	Class sums / ratios	50	247	7	2.5	2.75	0.024	1.3×10^{-2}
250 kb	Individual lipids	378	970	168	160.9	1.04	0.142	0.23
250 kb	Class sums / ratios	36	161	6	2.5	2.36	0.031	3.8×10^{-2}
500 kb	Individual lipids	297	656	162	157.2	1.03	0.205	0.29
500 kb	Class sums / ratios	28	121	5	2.7	1.83	0.035	0.13

branch (i), so the LCAT assignment was made from the reaction's specificity (RHEA:32843, RHEA:44236) and is marked LCAT* throughout. The substrates, products and RHEA identifiers of each branch are given in Supplementary Table S9 (reactions). These annotation-based hypotheses do not establish *in vivo* flux direction, and because CTL and LIN were sampled in different years, all LIN–CTL contrasts are genotype-matched between-trial ratios rather than causal effects of management.

For each branch, we computed a per-sample reaction-balance score as the mean log abundance of products minus substrates (Rose *et al.* 2023; Gieger *et al.* 2008; Illig *et al.* 2010), so a uniform multiplicative trial difference cancels. In genotype-

matched paired Wilcoxon tests across 355 genotype pairs, all four branches shifted, and all four kept their direction when re-computed on only the species measured in both trials. The results were positive for PNPLA3 (+0.315), LRO1 (+0.270), and LCAT* (+0.145), and negative for PNPLA1 (−0.256) (BH-adjusted $p < 10^{-52}$; Supplementary Table S9, reaction balance). The candidate genes supporting each branch are listed in Supplementary Table S9 (GWAS gene support). TG-producing branches shifted in one direction and the hydrolytic TG→DG branch in the other, showing that the class-level shift holds within genotypes, not just on average. Because substrates and products are different lipid classes, scale invariance does not remove

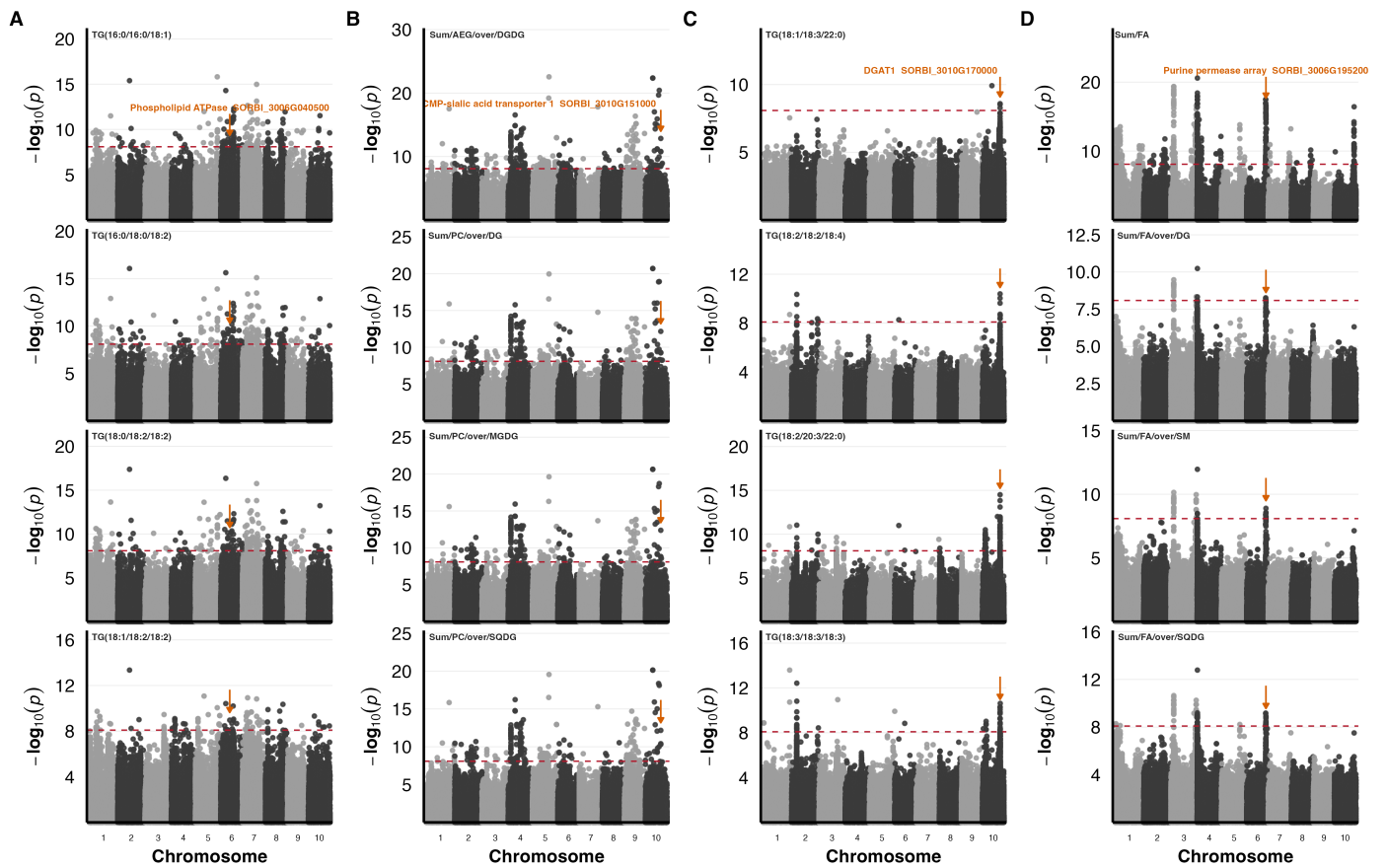


Figure 3 The same candidate locus recurs across multiple lipid phenotypes within each trial. Each column stacks four phenotypes that share one candidate locus. An orange arrow marks the lead SNP for that locus in every plot, and the candidate gene is named once, above the arrow in the top plot of the column. **(A)** CTL individual lipids, sharing SORBI_3006G040500 (phospholipid-transporting ATPase, chromosome 6). **(B)** CTL class sums and ratios, sharing SORBI_3010G151000 (CMP-sialic acid transporter 1, chromosome 10). **(C)** LIN individual lipids, sharing SORBI_3010G170000 (*DGAT1*, chromosome 10). **(D)** LIN class sums and ratios, sharing the chromosome 6 purine-permease array (SORBI_3006G195200). Within a column, each plot is identified by the trait named in its upper-left corner. Red dashed horizontal lines mark the Bonferroni genome-wide threshold ($p \leq 8.19 \times 10^{-9}$). Point shading alternates by chromosome. All Manhattan and QQ plots for the full phenotype set are browsable in the SoLD Shiny application.

Table 5 Per-class cross-trial overlap of individual-lipid GWAS candidates. Genes were assigned to a lipid class through the phenotypes with which they were associated. q , Benjamini–Hochberg-adjusted hypergeometric P -value. n.s., not significant.

Class	CTL	LIN	Shared	Fold	q
TG	866	890	37	1.63	9.4×10^{-3}
PC	58	1324	8	3.55	9.4×10^{-3}
Fatty acid	109	1897	7	1.15	0.95 (n.s.)
Terpenoid	37	39	0	—	1 (n.s.)
DG	13	34	0	—	1 (n.s.)
SQDG	9	396	0	—	1 (n.s.)
AEG	7	119	0	—	1 (n.s.)

class-specific trial artifacts. Thus, these directions reflect trial differences rather than management response.

Cross-referencing branches with the LIN GWAS candidate tables supported three of four (Supplementary Tables S6 and S9). Branch (i) was supported by two LCAT-superfamily genes, SORBI_3006G214500 and SORBI_3004G341900, whose closest Arabidopsis homologues are LCAT4 and PSAT; characterized members of this family release a lysophospholipid from PC or PE rather than transferring the acyl group to diacylglycerol (Chen *et al.* 2012; Banaś *et al.* 2005); branch (ii) by SORBI_3001G041900, a PNPLA-domain lipase in the SDP1 clade detected in both the individual-lipid and class-sum/ratio GWAS; and branch (iv) by SORBI_3010G170000 (*DGAT1*). No candidate was found for branch (iii), and no PDAT-family gene was recovered in either trial (Supplementary Table S9, branch summary). Together, these results link the class-level composition shift to four candidate reactions, three also showing a LIN genome-wide association.

An interactive SoLD Shiny application enables exploration of lipid and GWAS results

The SoLD web server provides an integrated environment for exploring lipid phenotypes and their genetic associations across two field environments (Fig. 6A). Traits can be examined at three

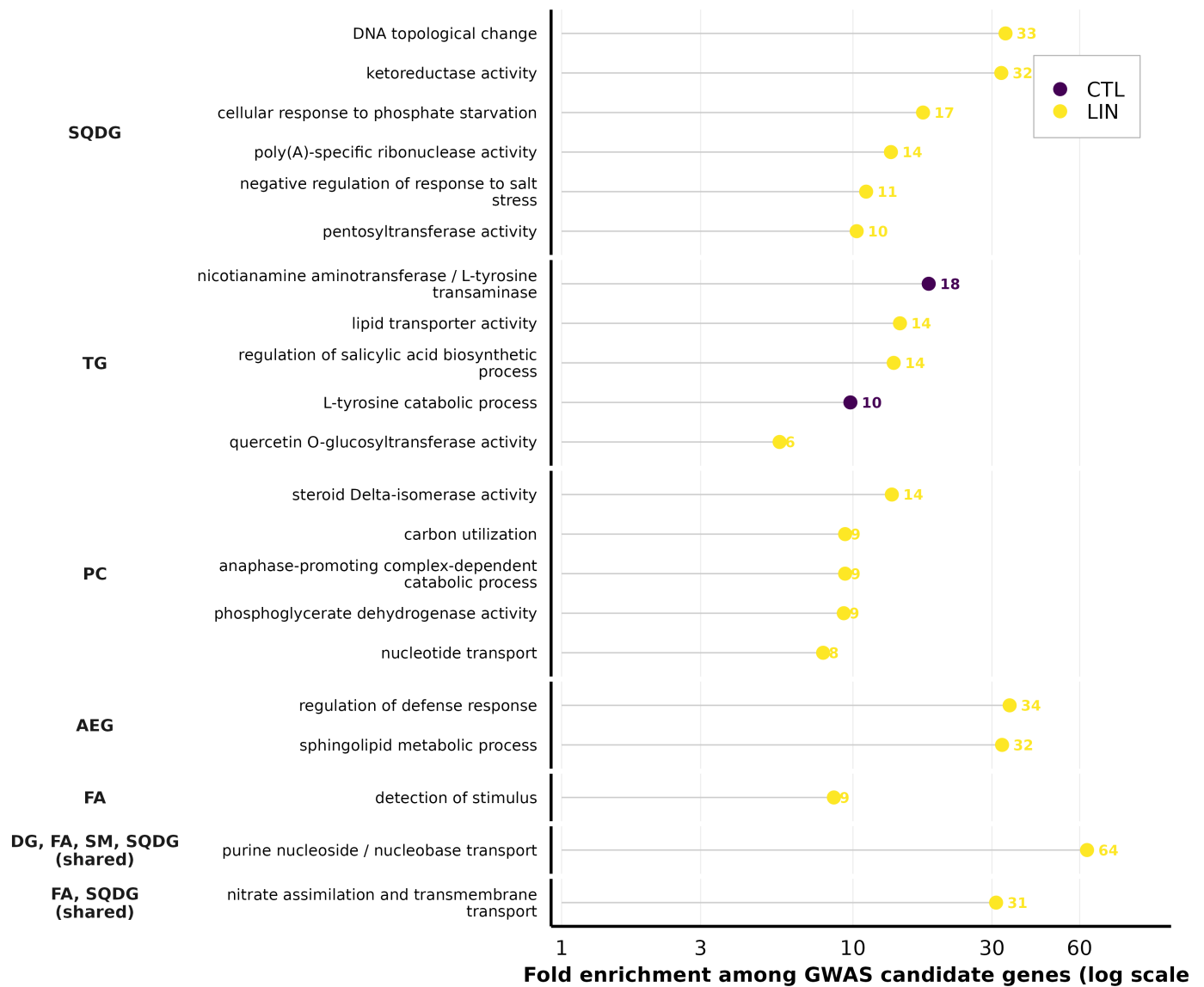


Figure 4 Gene Ontology terms enriched among LD-mapped GWAS candidate genes, grouped by the lipid class whose traits produced them. Nineteen loci are shown. Each began as an enrichment passing the LD-aware permutation null ($q_{LD} < 0.05$); terms sharing an identical candidate-gene set were then merged, first within a stratum, where the Gene Ontology's nested parent and child terms describe one locus several times, and then across lipid class, trait layer and ontology, where class sums and ratios share candidate genes by construction. Of the 50 loci remaining after those merges, the figure shows those whose GO term draws on at most 30 background genes, a criterion of interpretability rather than of significance: terms such as *zinc ion binding* (1,015 background genes) or *transmembrane transport* (526) pass the permutation null without naming a lipid-related process. Points give fold enrichment over chance on a logarithmic axis and are coloured by trial. Left-hand strips name the lipid class; the two strips marked shared are gene sets recurring across several classes, which are correspondingly less specific. The complete set of 50 loci, with the number of independent 250 kb intervals supporting each, is given in Supplementary Table S7 (loci collapsed).

resolutions: individual species, class-level sums, and inter-class ratios (Fahy *et al.* 2009) and hierarchical filtering by LIPID MAPS category, main class, and species. Exploratory and multivariate modules (data preview, boxplots, volcano plots, correlation and network views, heatmaps, PCA, t-SNE/UMAP) are paired with two genetic modules: (i) GWAS, which shows trait-specific associations, and (ii) Gene Hits, which aggregates recurrent candidate genes under user-defined dataset, trait-type, lipid-class, and significance settings. Furthermore, a Genotype module

supports experimental design by identifying accessions within specified lipid value ranges.

Also, the interface allows for easy tracing of traits to candidate genes through the combination of filtering, plotting, and gene summaries. In the *Gene Hits* module (LIN dataset; all trait sources; genome-wide threshold; strict LD, $r^2 \geq 0.9$), 1,325 candidate genes are output and ranked by the number of phenotypes in which they recur (Fig. 6B). The top gene is a chromosome 4 cytochrome P450 (S0RBI_3004G006400), recurring in 21 pheno-

A

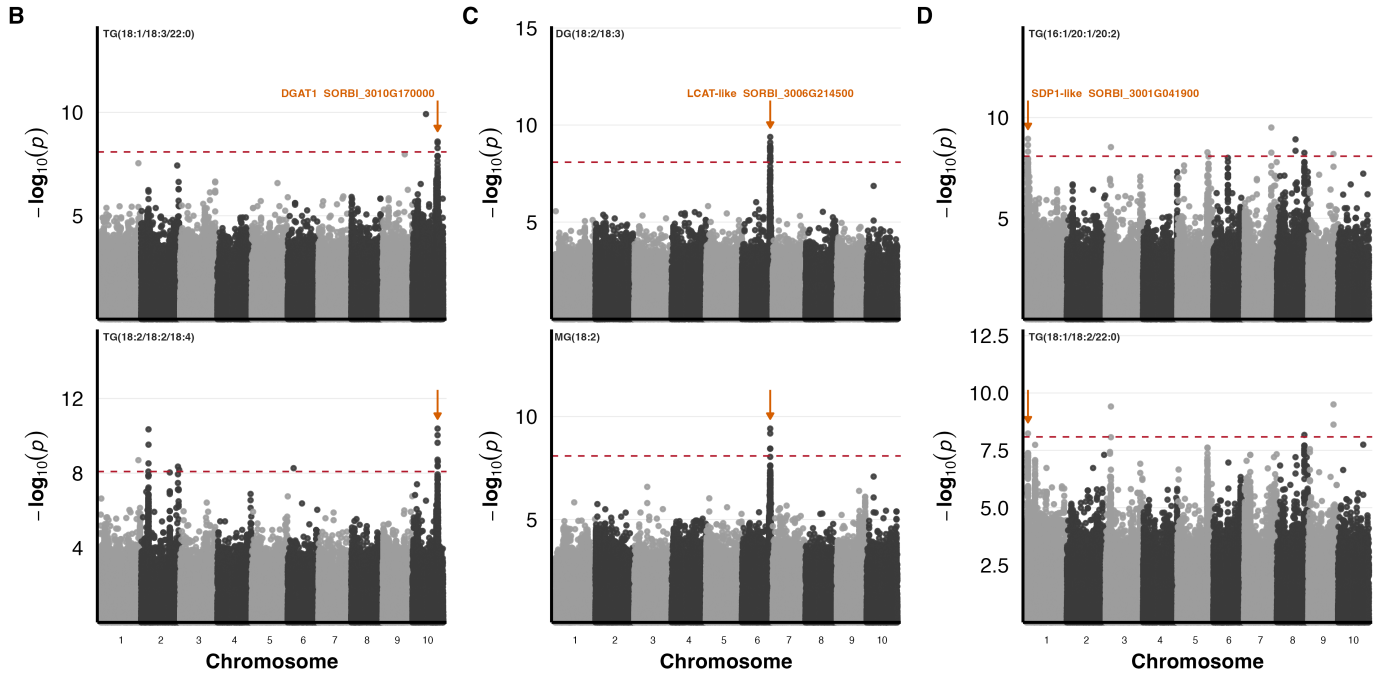
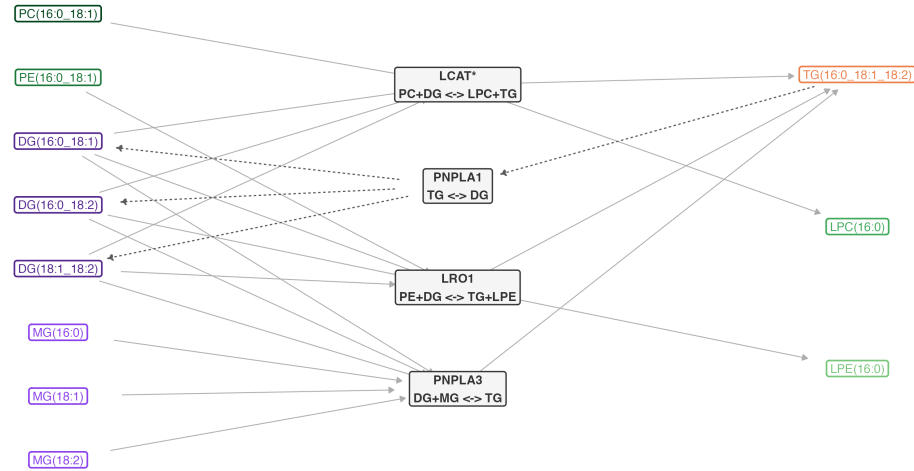


Figure 5 Candidate lipid reaction branches and the GWAS support for them. (A) LINEX-enriched lipid reaction subnetwork connecting candidate reactions to their substrate and product lipid species. Central reaction nodes depict hypothesized LCAT-like acyl editing, PNPLA1-mediated TG→DG lipolysis, LRO1-mediated acyl transfer between PE/DG and TG/LPE, and PNPLA3-associated DG/MG→TG re-esterification; arrows show reaction direction, dashed arrows run right to left, and boxed labels give representative lipid species coloured by class. All four reactions were retrieved from the LINEX2 network; the enzyme label on the acyl-editing branch, marked LCAT*, was assigned from reaction specificity rather than returned by LINEX2. **(B–D)** GWAS support for the three branch-supporting candidate genes under LIN, two associated traits per gene: **(B)** *DGAT1* (SORBI_3010G170000), **(C)** an LCAT-like gene (SORBI_3006G214500), and **(D)** an SDP1-like lipase (SORBI_3001G041900). An orange arrow marks the lead SNP at each candidate locus, and the gene is named once above the arrow in the top plot of its column; each plot is identified by the trait in its upper-left corner. Red dashed horizontal lines mark the Bonferroni genome-wide threshold, and point shading alternates by chromosome. The reaction-balance scores for these branches are given in Supplementary Table S9 (reaction balance) rather than shown here, because they are a CTL–LIN contrast.

types. Next are five adjacent chromosome 3 genes. Each carries a variant in perfect linkage disequilibrium ($r^2 = 1$) with the most significant SNP at the locus, recurs in 19–20 phenotypes,

and shares the same association ($p = 4.46 \times 10^{-20}$). These relate to one class sum (free fatty acids), one species (FA(22:1)), and 18 ratios. In the 500 kb window (7.40–7.90 Mb) around



Figure 6 SoLD Shiny app and the chromosome 3 candidate block under LIN. (A) The homepage of the SoLD Shiny app with a list of features. (B) Gene Hits module under the LIN dataset, all trait sources, the genome-wide threshold, and the strictest LD setting ($r^2 \geq 0.9$), ranking candidate genes by the number of distinct phenotypes each recurs in. (C) Regional view of the chromosome 3 locus that the top of that table points to, over 7.40–7.90 Mb for the LIN fatty-acid class sum. Top, every GWAS SNP as $-\log_{10}(p)$, coloured by its r^2 with the lead SNP, with the LIN Bonferroni threshold dashed and the lead SNP as a diamond; a second SNP at 3:7,739,242 ties it at $p = 4.46 \times 10^{-20}$. Middle, the 49 annotated genes in the window, coloured by the highest r^2 any variant inside them reaches with the lead SNP, which is how candidate genes are assigned throughout; grey genes carry no variant at $r^2 \geq 0.4$. Three of the five block genes are labelled, the other two being uncharacterised. Bottom, pairwise r^2 between variants less than 150 kb apart. All three tiers share one r^2 scale.

these genes, genome-wide significant SNPs and high-LD variants are restricted to ~7.70–7.78 Mb (Fig. 6C), and 28 of the 49 genes in the window were not selected ($r^2 < 0.4$). Three of the five block genes are well characterized. *SORBI_3003G088800* and *SORBI_3003G088900* are near-identical neighbors, closest to rice *OsLPR2* and *OsLPR1* (76% and 75% amino-acid identity), which encode multicopper oxidases of the LOW PHOSPHATE ROOT family that govern the root meristem response to phosphate deficiency. The third gene, *SORBI_3003G089000*, whose closest annotated Arabidopsis relative is *UBC24/PHO2*, acts as a negative regulator that limits phosphate uptake and root-to-shoot translocation. Two arms of the phosphate-starvation response, sensing at the root and restraint of transport, therefore fall within a single LD block detected under LIN. LD cannot distinguish among the five genes, and the associations are with fatty-acid ratios rather than phospholipid abundances. Thus, the SoLD Shiny app brings together lipid phenotypes, their associations, and LD-supported candidate genes, enabling direct recovery and inspection of loci (e.g., the chromosome 3 block), comparison of compositional and multivariate views across two trials, and selection of accessions by lipid content for follow-up experiments.

Discussion

The two trials differ in lipid composition at the class level. Lipid-trait loci, in contrast, were largely distinct between them. We first link the lipidome changes to trial conditions, then discuss candidate genes, and finish with the near-absence of shared loci between trials, which points to different genes acting on the same pathways in each environment. All GWAS results returned far more candidate genes than could be interpreted, with 1,100 for individual lipids and 115 for class sums and ratios under CTL, compared to 4,319 and 812 under LIN, because a single significant SNP tags every gene in its LD block. Two filters reduced these to the set discussed below. Most candidates fall within ontology terms passing the LD-aware permutation null, supported by both enrichment and association (Supplementary Table S7, genes in enriched terms). A second set comes from the LINEX reaction network, retaining genes mapped to branches flagged as remodeled between trials. These two approaches yield different types of candidates because enrichment clusters genes by function, while the reaction network maps them to the reactions connecting the lipid classes that showed changes. However, two caveats apply. First, gene functions are inferred by orthology to characterized genes in other species. Second, candidates come from LD blocks, so most associated regions contain several genes that should also be taken into account for validation.

Integrated interpretation of lipidome remodeling across contrasting field trials

The two most prominent changes in LIN are a decrease in SQDG and an increase in TG. The accumulation of TG is commonly associated with low temperatures, drought, oxidative stress, and nutrient-driven carbon reallocation, suggesting a shift towards increased storage within the LIN signature (Tan *et al.* 2018; Yan *et al.* 2025; Pfaff *et al.* 2020). The median LIN balances are higher than CTL for TG/DG (2.2 $\times$), TG/MGDG (2.7 $\times$), TG/SQDG (3.6 $\times$), and TG/PE (1.3 $\times$). The exception is TG/MG (0.74 $\times$), reflecting a faster rise in MG. DG shifts oppositely as DG/MG is 0.34 $\times$ and DG/PS is 0.13 $\times$. Correlations also change. PS–TG switches from negative in CTL ($r = -0.26$) to positive in LIN

($r = +0.43$), a strong sign reversal (Supplementary Table S4, CLR correlation delta). Additionally, LINEX reactions add to these results. TG-producing branches shift toward products under LIN, while TG→DG hydrolysis shifts toward substrate, collectively favoring TG over DG (Supplementary Table S9, reaction balance). These class-level balances together suggest a reorganization of DG supply and TG synthesis and mobilization, though direct measures of enzyme activity are needed. Both arms contain LIN candidates. *DGAT1* (*SORBI_3010G170000*) acylates diacylglycerol to triacylglycerol and falls within a repeatedly mapped chromosome 10 crude-fat QTL in sorghum (Boyles *et al.* 2017; Yen *et al.* 2008). The reverse (hydrolytic) step is *SORBI_3001G041900*, an SDP1-clade lipase. In Arabidopsis, SDP1 hydrolyses triacylglycerol to diacylglycerol and free fatty acid and accounts for most TG breakdown during reserve mobilization (Kelly *et al.* 2011). This locus clusters with *SbSDP1-L* in the sorghum lipase phylogeny (Sapara *et al.* 2024).

Another change involves the balance among phospholipid headgroups. PS increased against every class it was tested against (Supplementary Table S4, ratio statistics), and the PS–PC correlation flipped from positive in CTL ($r = +0.43$) to negative in LIN ($r = -0.34$), the largest shift among class pairs (Supplementary Table S4, CLR correlation delta). PE showed the largest increase in mean double-bond number of any class, although the species detected differ slightly between trials, and this is a composition measure rather than a direct account of acyl editing. Comparable shifts in headgroup proportions, membrane unsaturation, and lysophospholipid pools are reported under cold, nutrient, oxidative, and water limitation, where they accompany changes in membrane fluidity, lipid repair, signaling, and remobilization (Hou *et al.* 2016; Rawat *et al.* 2021; Yan *et al.* 2025; Zhang *et al.* 2020). The balance between the lysophospholipid classes also shifted, with LPC/LPE dropping to 0.50-fold in LIN (Supplementary Table S4, ratio statistics), driven by LPE rising from 0.0065% to 0.0139% of total signal while LPC was essentially unchanged (Supplementary Table S4, composition %TIC). PS and the lysophospholipids each contribute under 0.2% of TIC, so these are large proportional changes on small pools. The annotation is also shallow, with four LPC species under CTL and three under LIN and a single LPE species in both, so the shift cannot be read as evidence about phospholipase activity.

These changes do not indicate phosphate conservation, nor a headgroup swap. Under phosphate limitation, phospholipids are degraded to release phosphate and replaced by glycolipids, raising SQDG/DGDG and lowering phospholipids (Essigmann *et al.* 1998; Härtel *et al.* 2000; Jouhet *et al.* 2004; Kelly and Dörmann 2002; Pfaff *et al.* 2020), however, neither occurs here. On the CLR scale, SQDG drops most (-0.942 , $0.39\times$), galactolipids drop less (MGDG $0.77\times$, DGDG $0.85\times$), and PG shows no change (Supplementary Table S4), so the two anionic thylakoid lipids decline together rather than substitute (3.13% to 1.72% of TIC). Phospholipids hold steady (30.3% to 30.2% of TIC) while glycolipids fall (47.5% to 43.5%), increasing the phospholipid:glycolipid ratio. Additionally, all phospholipids increase relative to SQDG (PC $2.07\times$, PE $2.18\times$, PG $2.63\times$; Supplementary Table S4). PC salvage via deacylation would increase LPC, but LPC does not accumulate ($0.87\times$ CLR), so that route does not explain the modest PC decline. Overall, the lipidome suggests a smaller plastid lipid fraction with a disproportionate loss of SQDG. Because SQDG is the only sulfur-containing plastid lipid, this pattern is consistent with constrained sulfolipid synthesis. The LIN GWAS included two SQD2-like loci (*SORBI_3003G076900*, *SORBI_3002G000600*),

though they were implicated via fatty-acid traits, not SQDG. In the absence of sulfur, chlorophyll, or thylakoid measurements, and considering the other reported LIN vs CTL differences, the lipidome data alone constrain conclusions about the underlying mechanism. Thus, the LIN lipidome is best understood as a redistribution among plastid, phospholipid, lysophospholipid, and neutral-storage pools, supported independently only by the GWAS.

Genetic control of lipid variation is stronger under LIN and largely trial-specific

These compositional changes coincide with a shift in how strongly lipid variation reflects genotype. Under CTL, the median h^2 was near zero (11/163 species excluded zero), while under LIN the median h^2 was 0.170 (53/163 excluded zero), with h^2 higher under LIN for 131 species versus 21 under CTL. This difference persisted after conditioning on population structure and after removing any race group or 20% of accessions. Low heritability for individual leaf metabolites in a single field environment is usual rather than exceptional. Mean broad-sense heritability across 76 leaf metabolites in a 135-line wheat panel was 0.25, with genotype-by-location variance twice the genetic variance (?), and 25 to 40% of leaf mass features in a 282-line maize panel fell below $H^2 = 0.2$ (?). Our values are genomic heritabilities from a marker-based relationship matrix, so they capture only the additive variance tagged by markers and are expected to sit below broad-sense estimates from replicated designs. CTL is therefore better read as a low-genetic-variance environment. A similar one-sided pattern holds for ancestry, though effects are small. No lipid-class sum was associated with botanical race in either trial (smallest $q = 0.479$, $\epsilon^2 \leq 0.027$). No species was significant under CTL, whereas nine were under LIN, eight by marker-based genetic cluster alone and one by cluster and by race, led by β -sitosterol ($\epsilon^2 = 0.095$, $q = 7.7 \times 10^{-5}$) and PS(18:0/18:2) ($\epsilon^2 = 0.088$, $q = 1.2 \times 10^{-4}$). Thus, ancestry explains little in either trial, and a detectable signal appears only under LIN. Moreover, the GWAS mapping results show the same asymmetry, with traits that have higher heritability being simpler to map, so the longer LIN candidate list likely indicates higher h^2 rather than a greater number of genes.

Candidate loci were largely trial-specific. The two candidate sets overlapped more than chance ($2.39\times$), but a single association tags every gene in its linkage block, so one shared region is counted once for each gene it contains. Counting shared 250 kb regions instead of shared genes removes the excess for individual lipids, from 289 shared genes ($2.07\times$) to 168 shared regions ($1.04\times$, $p = 0.23$). Class sums and ratios are the exception, still enriched at $2.36\times$ ($p = 0.037$) when counted as regions, which makes class-level composites the most reproducible unit across trials. Sharing a region is also not the same as sharing a trait. For each of the 354 genes recovered in both trials, we compared the lipid classes with which it was associated in each, and only 90 (25.4%) had any class in common, while the remaining 264 were candidates in both trials but for different lipids (Supplementary Table S8, shared ranked). Because the trials differ by year, field, and batch, as well as nutrients, these results do not isolate a genotype-by-nutrient effect, and testing that would require repeating the nutrient contrast within a single year.

A cold-induced MADS regulator is associated with fatty-acid traits under LIN

Developmental timing and cold response might be studied on SORBI_3003G406800, a MADS-box transcription factor associated with fatty-acid traits under LIN for FA(22:1), PC(18:2_20:4), the fatty-acid class sum, and ten FA-to-class ratios. It carries the GO term *positive regulation of transcription by RNA polymerase II* in the SQDG class-sum and ratio layer. Multiple studies support its cold-response role. In sorghum, this gene is induced in cold, and it is a hub in the early shock co-expression module (Kurt 2026). Its functional homolog in maize, ZmMADS69, a flowering-time locus, regulates ZmRap2.7/ZCN8 and was selected during tropical-to-temperate adaptation (Liang et al. 2019).

LIN was planted earlier to expose the plants to cold during establishment, so a cold-induced gene linked to fatty-acid traits fits the treatment. Cold acclimation also involves acyl-chain remodeling to maintain membrane fluidity (Tan et al. 2018; Zhang et al. 2020). However, LIN also differs from CTL in nitrogen, phosphorus, and chemicals, so temperature cannot be isolated. A MADS-box flowering-time regulator may instead reflect a developmental-stage shift at sampling rather than lipid regulation, which this design cannot differentiate. Earlier planting also changes the photoperiod and temperature a plant experiences at each stage, so a flowering-time locus would be expected to differ between the trials, whether or not it acts on lipids. Thus, this locus is a strong candidate for earlier-planting, with cold response and developmental timing.

Tyrosine catabolism is enriched under CTL, and the dhurrin route is recovered under LIN

Candidates from both trials cluster around a single metabolic branch point on opposite sides. In sorghum, tyrosine is either catabolized via tyrosine aminotransferase (TAT) and 4-hydroxyphenylpyruvate dioxygenase (HPPD) to homogentisate and then to fumarate and acetoacetate, supplying acetyl-CoA for fatty-acid synthesis (Schenck and Maeda 2018), or diverted to dhurrin biosynthesis via CYP79A1, CYP71E1, and UGT85B1 (Busk and Møller 2002; Darbani et al. 2016). Because both pathways use the same precursor, candidates on either side suggest the trials differ in tyrosine allocation.

Under CTL, the *L-tyrosine catabolic process* is enriched, and all associated lipids are TGs. Two of the four genes carrying the term are consecutive enzymes of the pathway, the other two being nicotianamine aminotransferases (Inoue et al. 2008). SORBI_3005G152200 is the only tyrosine aminotransferase candidate and is associated with ten TGs, eight of which carry palmitate (16:0), and the next enzyme, 4-hydroxyphenylpyruvate dioxygenase (SORBI_3002G104200), is a candidate for a single lipid, TG(10:0_10:0_10:0). They sit on different chromosomes, so the term draws on independent loci rather than on one LD block. The pathway ends in fumarate and acetoacetate (Schenck and Maeda 2018), and acetoacetate is converted to acetyl-CoA, the carbon substrate for *de novo* fatty acid synthesis, with those fatty acids then esterified onto the glycerol backbone to form TG (Yen et al. 2008). Tyrosine catabolism therefore offers a plausible, if indirect, route from this locus to genotypic variation in TG content.

Under LIN, no tyrosine- or dhurrin-related GO term is enriched. Dhurrin is instead implicated by a single chromosome 1 locus that is strongly associated. The seven candidates share 12/13 PA-to-class ratios; three also carry the fatty-acid class sum and FA(22:1), and four carry TG(14:0_18:3_18:3).

Three candidates are known dhurrin-cluster genes (Darbani *et al.* 2016)—UGT85B1 (SORBI_3001G012400), SbMATE2 (SORBI_3001G012600), and SbGST1 (SORBI_3001G012500)—and a fourth, SORBI_3001G012000, is a CYP71 P450 annotated to the oxime monooxygenase step. The association does not extend to CYP79A1 (SORBI_3001G012300) or CYP71E1 (SORBI_3001G012200); although they lie between candidates, they are not tagged at $r^2 \geq 0.4$, indicating the signal covers only part of the cluster. Because all seven candidates share the same signal, the association maps a region rather than a specific gene. Only two candidates appear in any (broad) enriched term—iron ion binding (P450) and transmembrane transporter activity (transporter)—while UGT85B1 and SbGST1 appear in none. Thus, the dhurrin link is based on locus gene content and annotation, not enrichment.

Dhurrin is an alternative fate of tyrosine and functions as both a nitrogen store and a defense compound, making it relevant under reduced nitrogen. Nitrogen supply induces dhurrin synthesis in sorghum (Busk and Möller 2002), and its turnover releases nitrogen for early growth. High-dhurrin genotypes accumulate more shoot biomass than low-dhurrin genotypes under nitrogen deficiency (Emendack *et al.* 2025). Abiotic stress can also increase dhurrin independently of nitrogen. Salt stress upregulates tyrosine-metabolism genes and coincides with growth restriction and dhurrin accumulation (Amombo *et al.* 2026). Combined water, salinity, and heat stress doubles dhurrin while shifting the forage lipid profile (Giraldo-Acosta *et al.* 2026), and dhurrin contributes to pathogen defense (Fall *et al.* 2025). Under low inputs, reduced commitment of tyrosine-derived carbon and nitrogen to dhurrin, together with allelic variation in the dhurrin cluster, could change allocation and the precursor pool available for lipid synthesis.

Thus, under CTL, tyrosine catabolism is enriched across independent loci, while under LIN, the dhurrin route relies on cluster members within a single block. Both implicate the same branch point, so these data motivate follow-up on tyrosine allocation between catabolism and dhurrin rather than proving a difference.

Nitrate transporters are the remaining nitrogen-related candidates under LIN

Nitrogen transport enters the LIN candidate set through two GO enrichments. The first is specific but narrow. *Nitrate transmembrane transporter activity*, *nitrate assimilation*, and *nitrate transmembrane transport* are enriched in the FA and SQDG sets, and each rests on a single 250 kb interval on chromosome 4. The second GO term is broad but distributed, includes *Transmembrane transporter activity* and *transmembrane transport*.

The nitrate terms are driven by SORBI_3004G009400 (NRT2.1-like, high-affinity) together with two adjacent genes of the same family, SORBI_3004G009200 and SORBI_3004G009500, all three within a 250 kb interval on chromosome 4. SORBI_3004G009400 is the best supported of the three and is associated with FA(22:1) and with the FA/SQDG ratios. That interval tags 23 LIN candidate genes in total, so LD does not single out the transporters within it. In maize, NRT2.1 requires NAR2.1, and its capacity varies along the root with plasma-membrane H⁺-ATPase activity (Lupini *et al.* 2016; Sorgonà *et al.* 2011).

Two additional NPF/NRT1 transporters, SORBI_3008G081400 and SORBI_3001G133900, appear in the broad transport terms. SORBI_3008G081400 is recovered in both trials (seven palmitate-containing TGs under CTL;

FA(22:1) and TG(18:1_18:2_18:2) under LIN). Its closest Arabidopsis homolog is NPF4.6/NRT1.2/AIT1, which transports both nitrate and abscisic acid without the two competing as substrates (Kanno *et al.* 2013). The same protein interacts with PHOSPHOLIPASE D α 1 at the plasma membrane, and PLD α 1 is epistatic to NRT1.2 in ABA responses (Li *et al.* 2020b). The homolog therefore sits at a junction of nitrate transport, ABA transport and phospholipase D activity, which is a plausible point of contact between nitrogen status and membrane lipid turnover, although the PLD α 1 interaction was characterized in seed germination rather than in leaves. SORBI_3001G133900 associates with one lipid, TG(14:0_18:2_18:3), and its rice ortholog OsNPF2.4 mediates low-affinity nitrate root-to-shoot transfer (Xia *et al.* 2015). Thus, these three transporters are the most promising nitrogen-related candidates in the LIN set for follow-up work.

Phosphate and sulfur candidates recovered under LIN

Under LIN, the term cellular response to phosphate starvation is enriched in the SQDG class-sum and ratio traits, and its three genes are SORBI_3001G028500 on chromosome 1, and SORBI_3003G088800 and SORBI_3003G088900 on chromosome 3. SORBI_3001G028500 encodes an SPX-domain phosphate-status signaling protein and is associated with FA(22:1) and with the SQDG-to-fatty-acid ratios in both directions. Like rice OsSPX4, which negatively regulates phosphate signaling through its interaction with PHR2, it may regulate phosphate-starvation responses (Lv *et al.* 2014).

The two chromosome 3 genes sit in the strongest signal recovered under LIN, across 52 significant SNPs, associated with FA(22:1) and with 19 class-ratio traits. SORBI_3003G088800 and SORBI_3003G088900 are separated by 1.8 kb and are closest to the rice *OsLPR2* and *OsLPR1*, the multicopper oxidases of the LOW PHOSPHATE ROOT family, which control the root meristem response to phosphate deficiency. In rice, the family has expanded to five genes, *OsLPR1–5*, within 65 kb of chromosome 1, of which *OsLPR3* and *OsLPR5* are induced by Pi deficiency (?), so the sorghum pair is closest to the non-induced members. The same block carries a PHO2-family ubiquitin-conjugating E2, SORBI_3003G089000. PHO2/UBC24 is a negative regulator that limits phosphate uptake and root-to-shoot transport. The gene *pho2* overaccumulates Pi through a nonsense mutation in *UBC24*, and *PHO2* is miR399-cleaved under phosphate deficiency (?). PHT1 transporters and PHOSPHATE TRANSPORTER TRAFIC FACILITATOR1 act downstream of PHO2 (?), and in Arabidopsis the RING-type E3 ligase NITROGEN LIMITATION ADAPTATION partners with UBC24 to ubiquitinate PT2 (?), a link that matches our design because LIN reduced both N and P. The block holds five genes in all, the remaining two being a WD40-repeat protein (SORBI_3003G089100) and a glutamyl-tRNA amidotransferase subunit (SORBI_3003G089200), neither of which belongs to an enriched term. Because every gene in the block is tagged by the same signal, the association identifies the region rather than any one gene within it.

Two transporters are included via the candidate list rather than by enrichment, and both map only to the general term transmembrane transporter activity (not shown in Fig. 4). SORBI_3006G244000, a STAS-domain protein placed in the SULTR3 family by homology, is associated with 45 significant SNPs with PC(18:2_20:4) and two TG species. In Arabidopsis, SULTR3 proteins localize to the chloroplast envelope and mediate chloroplast sulfate uptake (Chen *et al.* 2019). Because

SQDG is a plastid lipid whose sulfur-containing head group depends on plastid sulfate import and reduction, SULTR3 sits on the sulfur-supply route to SQDG. Phosphate limitation typically raises SQDG by replacing phospholipids, yet SQDG was depleted under LIN, which makes constrained plastid sulfate supply a hypothesis worth exploring. SORBI_3002G116100 encodes a PHT1-family H⁺/Pi cotransporter whose closest characterized rice relative is OsPT4 (*OsPht1;4*), affecting phosphate accumulation and embryo phosphate mobilization (Ye *et al.* 2015; Zhang *et al.* 2015). Its five traits are four TGs and one PC rather than phospholipids, which does not fit a simple phospholipid-to-glycolipid replacement under phosphate limitation. Thus, these loci are prioritized for functional validation for LIN associated phenotypes.

A TIFY/JAZ region enriched under LIN is also recovered under CTL for different lipids

Under LIN, three TIFY-domain genes on chromosome 1 (SORBI_3001G259300, SORBI_3001G259400, SORBI_3001G259600) drive the enrichment of the regulation of defense response, regulation of the jasmonic acid mediated signaling pathway, and response to wounding, all from the AEG trait class. All three associate with AEG(0-15:2_15:0) and PC(14:0_18:3), and SORBI_3001G259400 and SORBI_3001G259600 additionally associate with 11,14,17-eicosatrienoic acid. The same region is also recovered under CTL, but for different lipids and without a corresponding enrichment. There, SORBI_3001G259400 and SORBI_3001G259600 are candidates for the nine TG species, seven of which contain 16:0. SORBI_3001G259700 is a candidate for four of those nine, and none of the genes is a candidate for any TG under LIN.

TIFY/JAZ proteins repress jasmonate signaling, and jasmonate slows growth while redirecting metabolism toward defense and repair (Larrieu and Vernoux 2016). Jasmonate is made from α -linolenic acid (18:3) released from plastid galactolipids, and the LIN traits include PC(14:0_18:3), which carries that precursor, and a free C20:3. Nitrogen, phosphorus, and temperature all differ between CTL and LIN, and each has a documented connection to jasmonate signaling. Nitrate signaling can act through NIN-LIKE PROTEIN 7 to activate JAZ1, while jasmonate can restrain nitrate-driven growth through COI1-dependent JAZ1 degradation (Pu *et al.* 2025). Phosphate deficiency can also increase jasmonate and jasmonoyl-isoleucine levels and enhance herbivory resistance (Khan *et al.* 2016). Low temperature provides another relevant stress context, as most *OsTIFY* genes respond to cold in rice (Ye *et al.* 2009), and TIFY-family genes have also been implicated in cold responses in pepper and sweet orange (Wang *et al.* 2023; Zhang *et al.* 2026). The whole pathway is a plausible point of contact for future studies. However, LD does not isolate the TIFY genes. The same significant SNPs also tag an SPX domain-containing protein (SORBI_3001G259100, explained above), a glycosyltransferase, a translation initiation factor, and several uncharacterized genes in the same block. The whole region might be of interest for future studies.

Conclusion

SoLD is a population-scale lipidomics web resource for SAP, profiling 266 annotated lipid species across two contrasting fields. Compared with CTL, the LIN lipidome showed four population-wide shifts: SQDG depletion, TG enrichment, a PS-centered phospholipid redistribution, and an LPC/LPE balance shifted toward LPE. These were independent of botanical race

and genome-wide genetic structure. Genomic heritability was also higher under LIN with a median near zero under CTL compared to 0.170 under LIN. The increase reflects more genetic variance rather than less experimental noise, so more of the lipidome can be mapped under LIN than CTL.

GWAS returned far more candidate genes than could be interpreted. Under CTL we obtained 1,100 for individual lipids and 115 for class sums and ratios, and 4,319 and 812 under LIN. Two filters reduced these numbers based on ontology terms, passing an LD-aware permutation null and genes mapped onto LINEX reaction branches. The two filters return different kinds of candidates. Enrichment groups genes by function, whereas the reaction network assigns them to the specific reactions linking the lipid classes that changed between trials. Both arms of DG-TG cycling are recovered by this route, including SORBI_3010G170000 (*DGAT1*) for polyunsaturated TGs and an SDP1-like lipase for the reverse reaction. GO enrichment identifies nutrient and cold candidates. SORBI_3006G244000, a SULTR3-family chloroplast sulfate transporter, offers a potential mechanism for the SQDG decline that the canonical phosphate-starvation response does not explain. Furthermore, the strongest recurrence under LIN is a chromosome 3 block carrying two LOW PHOSPHATE ROOT homologues, which share an enriched *cellular response to phosphate starvation* term with an SPX-domain signaling gene on chromosome 1. Additionally, nitrogen is represented by SORBI_3004G009400, a high-affinity nitrate transporter driving nitrate transport enrichment in both the fatty-acid and SQDG trait sets. The loci are largely trial-specific, but the implicated metabolism is shared. Tyrosine is the clearest case, with the catabolic route enriched under CTL and the dhurrin cluster, sorghum's cyanogenic glucoside and nitrogen store, recovered under LIN, while the same TIFY/JAZ region appears in both trials with different associated lipids. Leaf-lipid genetic control is, therefore, largely environment-specific at the locus level, with the few pathways recovered in both trials. Class-level composite traits are the more reproducible unit across environments. Thus, these are candidates, not causes, and disentangling nutrient from year and planting date effects will require factorial, same-year trials including both regimes.

Materials and methods

Plant Material and Growth Conditions

We used the Sorghum Association Panel (~400 genotypes), a genetically diverse collection of sorghum accessions representing major botanical races and broad phenotypic diversity (Casa *et al.* 2008; Boatwright *et al.* 2022). We evaluated SAP accessions across two different field settings during two growing seasons (2019 and 2022) at the Pee Dee Research and Education Center, Clemson University, Florence, South Carolina. The “control” condition, herein denoted as CTL, involved standard agronomic inputs with sufficient levels of nitrogen (N) and phosphorus (P) along with a typical planting schedule. In contrast, the “low input” condition, denoted as LIN, featured reduced N and P coupled with earlier planting to mimic a cold stress environment.

Schematic diagram

The schematic in Supplementary S1 Fig was created using FigureLabs.ai (<https://www.figurelabs.ai/>).

Lipidomics Analysis

Sample Preparation and Extraction. Samples were prepared following the standard extraction protocols explained in the paper by (Barnes *et al.* 2022).

Liquid Chromatography-Mass Spectrometry (LC-MS) Analysis. Lipid extracts were analyzed at the Joint Genome Institute/Lawrence Berkeley National Laboratory using their standard LC-MS/MS electrospray ionization lipidomics workflow; detailed chromatographic conditions, acquisition parameters, and quality-control procedures are described in the published JGI/LBNL protocol using a Thermo Exploris 120 Orbitrap mass spectrometer inline with an Agilent 1290 UHPLC system (Louie *et al.* 2025).

Feature Detection and Data Processing. Raw LC-MS data, in both positive and negative ion modes, were processed utilizing MZmine 2, an open-source software for the analysis of mass spectrometry data (Pluskal *et al.* 2010). The pipeline encompassed peak detection, chromatogram construction, deconvolution, and isotope filtering, producing a detailed feature table containing mass-to-charge ratio-retention time (mz-rt) pairs. Isotopic peaks were excluded to minimize redundancy, as a single metabolite can yield multiple co-eluting ions, such as adducts and in-source fragments; remaining mz-rt duplicates were resolved at the annotation stage described below. We acknowledge that such degeneracy can lead to an inflated number of features compared to the actual number of metabolites present, which we considered during metabolite identification.

Blank/extraction-control filtering, intensity thresholds, and sparsity pruning. To reduce background and carryover effects, an extraction control filter was implemented at the feature level. For each feature, the maximum intensity was determined across extraction controls (*a*) and biological samples (*b*). Features for which $b < 10 \times a$ were eliminated. To preclude the exclusion of borderline yet potentially biological signals, a feature was retained if at least one biological sample exceeded the extraction control maximum. Furthermore, a minimum average intensity threshold within the treatment groups of interest ($\sim 10^6$ peak height) was imposed to ensure that downstream analyses would emphasize robust signals. At the sample level, any sample exhibiting $\geq 70\%$ features as zero (or missing) was excluded prior to normalization and statistical analysis. This pruning of sparsity is essential to prevent unstable scaling and spurious differential signals caused by ultra-sparse profiles. A lipid species was considered detected within a condition if it passed the feature-detection, intensity, and sparsity filters described above using only that condition's samples. Species detected in both conditions were classified as shared, whereas species detected in only one condition were classified as condition-specific (CTL-specific or LIN-specific).

MS/MS spectral library matching and cross-referencing of IDs. For each feature analyzed by MS/MS, the most intense fragmentation spectrum was queried against the GNPS database. Library matches resulted in putative identifications (Metabolomics Standards Initiative levels 2/3), potentially including isomers or near-mass analogs. Species-level assignments are therefore putative throughout, and individual annotations should be confirmed against authentic standards before being treated as identified compounds. Features without direct matches were eliminated. To enable quantification with identifications, tables were linked using the feature *row ID* from the MZmine peak list and the

corresponding #Scan# key in the GNPS results, ensuring a one-to-one correspondence between intensities and candidate identifications. In cases where a feature yielded multiple GNPS hits, a single primary annotation was designated by retaining the highest MQScore (cosine similarity). Ties in the values were resolved based on a greater number of shared fragment ions and a smaller precursor mass error (ppm). All other sub-threshold or lower-ranked candidates were retained for verification but were excluded from subsequent statistical analyses.

Systematic Error Removal Using Random Forest (SERRF). Following the cleaning process, the data were then used for SERRF normalization (Fan *et al.* 2019). We used the SERRF server (<https://slfan2013.github.io/SERRF-online/>) to obtain the normalized output. After applying SERRF, only biological samples were preserved. Any zeros were substituted with two-thirds of the minimum nonzero value for that feature to prevent potential infinite logarithmic transformations.

Spatial Correction. Finally, we conducted an additional quality control step specifically aimed at eliminating any spatial patterns across our experimental trials. This was achieved using the R package SpATS (Rodriguez-Alvarez *et al.* 2018), which applies a two-dimensional P-spline ANOVA surface over the field coordinates. For every lipid feature, we characterized its intensity as

$$y_{ij} = \mu + f_{\text{row}}(i) + f_{\text{col}}(j) + f_{\text{row,col}}(i, j) + \varepsilon_{ij}, \quad (1)$$

where

- f_{row} and f_{col} represent smooth functions that model systematic effects across rows and columns, respectively,
- $f_{\text{row,col}}$ is a smooth interaction surface that handles more complex spatial gradients.

For downstream analyses we used SpATS-adjusted intensities, calculated as the model residuals plus the fitted mean. This preserves the original intensity scale and avoids negative values, which is required for the %TIC, CLR, and log-ratio transforms. This methodology effectively corrects for positional artifacts, such as edge effects, that could interfere with subsequent analyses. Detailed smoothing parameters, including the number of knots, penalty orders, and comprehensive model specifications, can be found in our GitHub repository.

Lipid Quantification. The identified lipid species were assigned to the eight LIPID MAPS categories, and to classes and subclasses within them, following the LIPID MAPS classification (Fahy *et al.* 2009) (Supplementary Table S10). Category is the top tier used in the figures; lipid class refers throughout to the abbreviation carried in a species name, such as PC or TG. In each sample, the total intensity for a class was obtained by summing the intensities of the species within that class. To manage variability in signal intensity due to different runs or injections, these class totals were normalized relative to the total ion current (TIC) of the sample. As a result, the relative abundances were presented as percentages of the TIC by adding up intensities across all lipid classes in the sample. For each class and its subclasses, we determined the TIC fraction for each sample and then averaged these percentages across samples for each condition. Lipids were categorized into glycerolipid, glycerophospholipid, sphingolipid, sterol, betaine lipid, fatty acid, ether lipid, and terpenoid.

Lipid ratio calculation. For each sample i and lipid class c , species intensities were aggregated to class totals,

$$C_{i,c} = \sum_{k \in c} I_{i,k}.$$

Two annotated features were excluded from all class- and category-level comparisons. Phytosphingosine rose from 0.04% of total signal under CTL to 16.9% under LIN, which would make a free sphingoid long-chain base the second most abundant feature of the LIN lipidome and is not biologically plausible for a leaf. SM(d18:1/17:0) was the only member of its class, and plants do not synthesize sphingomyelin, so an odd-chain C17 species of this type is most consistent with an internal standard or misannotation. Both differences track the separate 2019 and 2022 analytical runs rather than genotype. Neither feature yielded a GWAS candidate, so no genetic result depends on this exclusion. Because log-ratios are scale-invariant, TIC normalization is not required for ratio computation. To handle zero class totals, we used a sample-specific pseudo-count,

$$\varepsilon_i = 0.5 \cdot \min\{C_{i,c} : C_{i,c} > 0\},$$

and computed

$$\ell_{i,c} = \log_{10}(C_{i,c} + \varepsilon_i).$$

Pairwise class log-ratios were then

$$r_{i,c_1/c_2} = \ell_{i,c_1} - \ell_{i,c_2}.$$

Compositional transforms. Class totals and the pairwise class log-ratios defined above are the traits carried forward to GWAS. For species-resolved analyses we used the centered log-ratio (CLR) transformation, which rescales each species by the geometric mean of all species in the same sample and so removes the constant-sum constraint of compositional data:

$$\text{CLR}(x_i) = \ln\left(\frac{x_i}{g(\mathbf{x})}\right),$$

where $g(\mathbf{x})$ is the geometric mean of species-level abundances in a sample.

Lipid species inventory

Species counts report every annotated feature in a trial, whether it is named in lipid shorthand, the convention that names a species by its class and acyl composition, or under a trivial chemical name, as the carotenoids, sterols, tocopherols, quinones and free sphingoid bases are. The same set is used for the species counts and for every composition analysis, so the %TIC category values cover it in full. Class labels follow the lyso normalisation used throughout, so a species written with an empty second chain is counted as the lysolipid rather than as its diacyl parent. The frozen species set is given in Supplementary Table S3.

Principal Component Analysis (PCA)

Principal component analysis (PCA) was performed on individual lipid species within each experimental condition. Lipid intensities were first normalized to total ion current (TIC) to obtain relative abundances, then transformed using a centered log-ratio (CLR). The CLR-transformed data were mean-centered and scaled to unit variance, and PCA was conducted in R using the *stats* package. The first two principal components were retained for visualization.

Genome-wide Association Studies (GWAS)

GWAS analyses were carried out for each lipid trait under each condition using the mixed linear model (MLM) featured in GEMMA (v2.3) (Zhou and Stephens 2012). To address population structure and relatedness, a centered relatedness matrix (kinship) was computed from SNP genotype data. We filtered the genotype data by MAF 0.05 and heterozygosity of 0.8. For each lipid trait, the MLM was applied using the kinship matrix to handle population stratification effects. Besides individual traits, GWAS was also applied to summed lipid classes and all possible ratios. The first three genotype principal components were fitted as fixed covariates. A Bonferroni-corrected genome-wide significance threshold was employed to account for multiple comparisons. Based on approximately 6.11 million tested SNPs under CTL and 6.08 million under LIN, this corresponded to $p \leq 8.19 \times 10^{-9}$ for CTL ($-\log_{10}(p) \geq 8.09$) and $p \leq 8.22 \times 10^{-9}$ for LIN ($-\log_{10}(p) \geq 8.09$).

Botanical race and population-structure testing

Botanical race, working group, and marker-based genetic cluster assignments for the SAP were taken from panel passport data. For race analysis, accessions were grouped into the five classical races (Bicolor, Caudatum, Durra, Guinea, Kafir) plus a Mixed group containing all hyphenated intermediate designations; accessions lacking a race assignment and the single *S. bicolor* subsp. *verticilliflorum* accession were excluded. The second grouping used the $K = 6$ genetic cluster assignment. Fourteen traits were tested per condition: total lipid signal ($\log_{10}$ of the summed SpATS-adjusted intensity across all annotated lipid features) and the %TIC of each of 13 major lipid classes. Groups with fewer than five accessions in a condition were excluded from that test. Differences among groups were assessed with tie-corrected Kruskal–Wallis tests within each condition, and P -values were adjusted across the 14 traits per condition using the Benjamini–Hochberg procedure. Effect sizes were computed as $\varepsilon^2 = (H - k + 1) / (n - k)$, where H is the tie-corrected statistic, k the number of groups, and n the number of accessions; negative values were truncated to zero. Because total ion current is partly technical, the total-lipid trait is reported for completeness and interpreted only relatively.

The stability of the class composition within a trial was assessed by resampling rather than by hypothesis testing. Neither procedure poses a null hypothesis, computes a test statistic, or yields a P -value. The first is a leave-one-group-out sensitivity analysis: within a trial, every accession belonging to one botanical race group was deleted and the mean %TIC of each of the 13 lipid classes was recomputed from the remaining accessions, once for each of the six groups, giving 78 recomputations per trial. We report the smallest and largest value any omission produced and the largest absolute departure from the full-panel mean. The second is nonparametric subsampling: 1000 draws of 80% of a trial's accessions were taken without replacement (315 of 394 under CTL, 290 of 363 under LIN, random seed fixed at 1), the same 13 class means were recomputed for each draw, and the 2.5th and 97.5th percentiles of the resulting distribution are reported as a percentile interval. That interval describes how far the descriptive mean moves under resampling; it is not a confidence interval derived from a test, and it carries no significance claim. Both procedures run entirely within a single trial and involve no comparison between trials.

Candidate-locus definition and cross-trial overlap testing

Cross-trial overlap was computed from the LD-mapped candidate tables, separately for the individual-lipid layer, the class-sum/ratio layer, and their union. At gene level, the background was the 34,027 annotated *S. bicolor* v3.1 gene models. Because LD-based mapping assigns every gene within a window to one association, gene-level counts overstate independent signals, so we repeated the analysis at locus level. The genome was partitioned into non-overlapping 250 kb windows, and a window was a candidate locus for a condition if it contained at least one candidate gene from that condition. The locus-level background was the set of windows containing at least one annotated gene (2,279 windows). At both resolutions we assessed overlap with a one-sided hypergeometric (Fisher right-tail) test, and reported P -values with the fold enrichment over the expected overlap $|A||B|/N$ and the Jaccard index $|A \cap B|/|A \cup B|$, which is insensitive to set-size disparities between conditions.

For the per-class analysis, each candidate gene was assigned the lipid classes represented among the phenotypes with which it was significantly associated. Class labels were parsed from the lipid species name for individual-lipid traits and from the numerator and denominator of the trait name for class sums and ratios; species lacking an abbreviated class prefix were mapped via the curated lipid class table. Because a single gene in the class-ratio layer can be associated with ratios spanning many classes, per-class overlap tests were stratified by trait layer. Class concordance was the proportion of genes recovered in both conditions that shared at least one lipid class between their CTL and LIN class sets. All tests were implemented in Python (pandas, NumPy), with hypergeometric and Benjamini–Hochberg procedures implemented directly.

Gene Annotation

SNPs were aligned with the *Sorghum bicolor* reference genome v3.1 (BTx623). Linkage disequilibrium in the sorghum association panel extends over appreciable physical distances (Boatwright *et al.* 2022), so a Bonferroni-significant SNP typically tags several linked genes rather than one. Significant SNPs were therefore grouped into association regions at a linkage-disequilibrium threshold of $r^2 \geq 0.4$. This is a moderate cutoff, retaining variants that carry at least approximately 40% of the effective sample size for tagging a causal site, and it trades the narrow regions obtained at stringent thresholds, which discard genes tagged by imperfect proxies, against the inflated gene counts obtained at permissive ones. Because candidate-gene counts are considerably more sensitive to this cutoff than interval counts are, all interval-level analyses, including GO enrichment and the cross-trial overlap, were carried out on 250 kb intervals rather than on gene identifiers. Candidate genes were defined as the annotated gene models falling within these LD-defined association regions. Functional annotations and homology were obtained from Phytozome (<https://phytozome.jgi.doe.gov>), SorghumBase (<https://sorghumbase.com>), and TAIR for corresponding *Arabidopsis thaliana* orthologs. Genes with known roles in N, P, cold tolerance, or lipid metabolism were specifically noted. We aggregated the frequency of each candidate gene within all lipid GWAS findings and marked those with the highest recurrence among Bonferroni-significant associations.

Gene Ontology Enrichment GO biological-process and molecular-function terms were taken from the same PANTHER-derived annotation. The two ontologies were analyzed separately, each against a background restricted to sorghum

genes carrying at least one annotation in that ontology. Overrepresentation was tested with a right-tailed hypergeometric test, retaining terms with at least three candidate genes and five background genes, and p -values were BH-corrected across all GO terms within each ontology and analysis. To account for LD among candidate genes, each term was also tested against a permutation null that preserved the candidate block structure. We used 8,000 draws, which bounds the smallest attainable permutation p -value at 1.25×10^{-4} , well below the resolution needed after BH correction. The genome was divided into consecutive, non-overlapping 250 kb intervals, assigning each candidate gene to the interval containing its start, so the number of intervals carrying a term reflects the number of distinct genomic regions supporting that term rather than the number of gene IDs. Each permutation redrew the same number of intervals, each contributing the same number of genes, from background intervals matched on gene density. Permutation p -values were BH-corrected and reported as q_{LD} .

Enrichment was computed separately within each lipid class, so that a term is attributed to the traits of one class rather than to the candidate set as a whole. Crossed with the two trait layers, the two ontologies and the two trials, this gives one analysis per class per layer per ontology per trial.

Because the Gene Ontology nests parent and child terms, a single candidate gene set can surface as several terms. Terms whose candidate gene sets were identical were therefore merged into one finding, keeping the most specific term as the label and retaining every merged term name and GO identifier in the supplementary table. The same gene set also recurs between analyses, since class sums and ratios share candidate genes by construction. Gene sets identical across analyses were merged in the same way, and the lipid classes in which each appeared were retained, because a gene set surfacing in a single class is class-specific whereas one surfacing across many is generic. After both merges, the 101 terms passing the LD-aware null reduced to 50 distinct candidate gene sets.

We retained gene sets whose GO term draws on at most 30 background genes. This is a criterion of specificity rather than of significance, applied because a term drawn from a background of hundreds or thousands of genes cannot be interpreted whatever its enrichment. *Zinc ion binding* (1,015 background genes), *transmembrane transport* (526), *nucleic acid binding* (522) and *mRNA binding* (432) all pass the permutation null, and none of them names a lipid-related process. Twenty-one gene sets remain and are shown in Fig. 4. All 50 are listed in Supplementary Table S7 together with the number of independent 250 kb intervals supporting each, and the uncollapsed per-term results. Interval count is reported but was not used to select or to rank terms, because it scales with the size of the GO term and a threshold on it would discriminate against precisely the small, specific terms of interest. Terms supported by three or more intervals averaged 175 background genes at 10-fold enrichment, while those below that threshold averaged 24 background genes at 23-fold.

Genomic heritability

Genomic heritability was estimated as

$$h^2 = \frac{V_g}{V_g + V_e}$$

from the model

$$y = \mathbf{1}\mu + u + e,$$

where

$$u \sim N(0, K\sigma_g^2)$$

and K is a genomic relationship matrix. The matrix K was built with PLINK 2 using `-make-rel square` from the SAP genotype set used for GWAS after LD pruning with `-indep-pairwise 50 5 0.2`, covering 400 genotypes. Variance components were estimated by REML using the eigendecomposition of K .

Intervals are 95% profile-likelihood intervals, defined as the set of h^2 values on a grid whose restricted log-likelihood falls within $\chi_1^2/2$ of the maximum. A trait was called non-zero when this interval excluded zero. Bootstrap resampling was not used, as resampling genotypes with replacement creates pairs with relationship 1 and identical phenotypes, which forces the estimate to 1 in every replicate. CTL and LIN were fitted using the same 351 genotypes and the same 163 species present in both trials, so the trials differed only in the phenotype. Both trials were also analysed jointly. Values were standardised within trial, and each genotype's mean across trials and difference between trials were fitted with the same model and the same interval, so that the heritability of the mean estimates the genotype main effect and the heritability of the difference estimates genotype-by-environment variance. Because the trials differ in year, field, planting date and analytical batch and neither is replicated, the environment term is the whole trial contrast rather than the nutrient treatment.

Lipid Metabolic Network Analysis (LINEX2)

We used the Lipid Network Explorer (LINEX2; <https://exbio.wzw.tum.de/linex/>) to reconstruct annotation-guided lipid reaction networks and identify plausible biochemical connections among lipid classes and candidate enzymes. Lipid names were standardized to align with species notation (class plus acyl composition). LINEX2 constructs a global species network based on curated reaction rules, including headgroup interconversions, (de)acylation/editing, elongation, and desaturation, and maps these relationships onto the observed lipid abundance data. Since *Sorghum bicolor* is not implemented as a default organism in LINEX2, we selected *Oryza sativa* (OSA) as the reference species for network construction.

For branch-level analysis, we extracted candidate reactions linking DG/MG, TG, and lysophospholipid pools and summarized each branch using a reaction-balance score. Three were taken from the LINEX2 network. The fourth, an LCAT-like acyl-editing branch denoted LCAT*, was defined manually from reaction specificity (RHEA:32843, RHEA:44236) rather than retrieved from LINEX2, and the asterisk marks it as such throughout. For sample i and lipid class c , let $C_{i,c}$ denote the class-sum abundance of class c in sample i , let ε_i denote the sample-specific pseudocount added prior to log transformation, and define $\ell_{i,c} = \log_{10}(C_{i,c} + \varepsilon_i)$ as the corresponding log-abundance. For each candidate reaction, we computed

$$S_{i,\text{rxn}} = \text{mean}(\ell_{i,\text{products}}) - \text{mean}(\ell_{i,\text{substrates}}),$$

so that larger $S_{i,\text{rxn}}$ indicates a more product-weighted balance within that sample, whereas smaller values indicate a more substrate-weighted balance. The branch-specific scores were defined as:

- LCAT*: $S_i = \text{mean}(\ell_{i,\text{LPC}}, \ell_{i,\text{TG}}) - \text{mean}(\ell_{i,\text{PC}}, \ell_{i,\text{DG}})$;
- PNPLA1: $S_i = \ell_{i,\text{DG}} - \ell_{i,\text{TG}}$;
- LRO1: $S_i = \text{mean}(\ell_{i,\text{TG}}, \ell_{i,\text{LPE}}) - \text{mean}(\ell_{i,\text{PE}}, \ell_{i,\text{DG}})$;
- PNPLA3: $S_i = \ell_{i,\text{TG}} - \text{mean}(\ell_{i,\text{DG}}, \ell_{i,\text{MG}})$.

These scores were treated as branch-aligned summaries of lipid-

pool balance rather than direct measures of enzymatic activity or metabolic flux. Reaction-balance scores were compared between CTL and LIN using paired genotype-matched Wilcoxon signed-rank tests, and p -values were adjusted using the Benjamini-Hochberg procedure.

Lipid Ontology Enrichment and Hierarchical Classification (LION/web)

We conducted a functional ontology for lipids using LION/web (Molenaar *et al.* 2019) (<http://www.lipidontology.com>). Lipid nomenclature was standardized according to LIPID MAPS annotations. The default settings of LION/web were applied for enrichment statistics and multiple-testing correction. We retained terms for $q \leq 0.10$. A hierarchical classification analysis of individual lipids and functional ontologies was also performed using LION/web.

Data Availability

Data processing and statistical analyses were performed in R (version 4.3.3). All the codes, figures, and pipeline are described in the GitHub repository: <https://github.com/rr-lab/SAP-Lipidomics-Database>.

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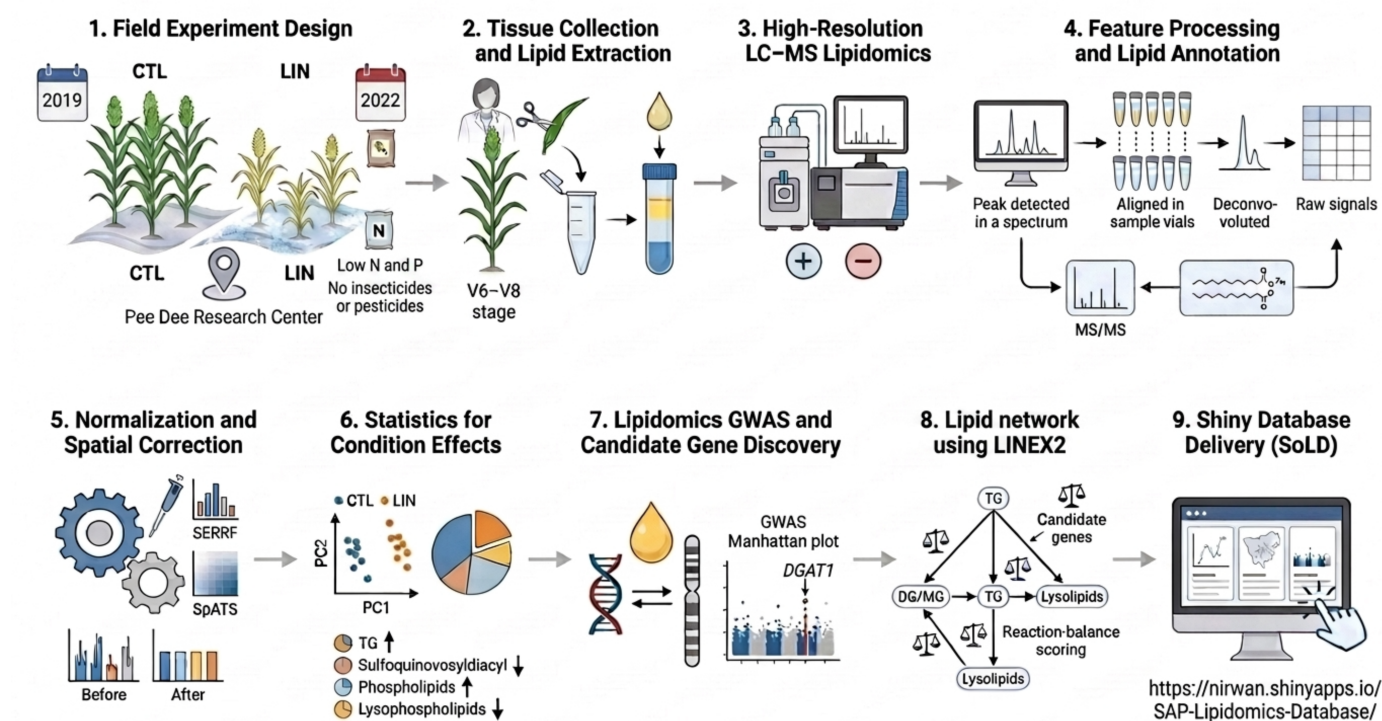
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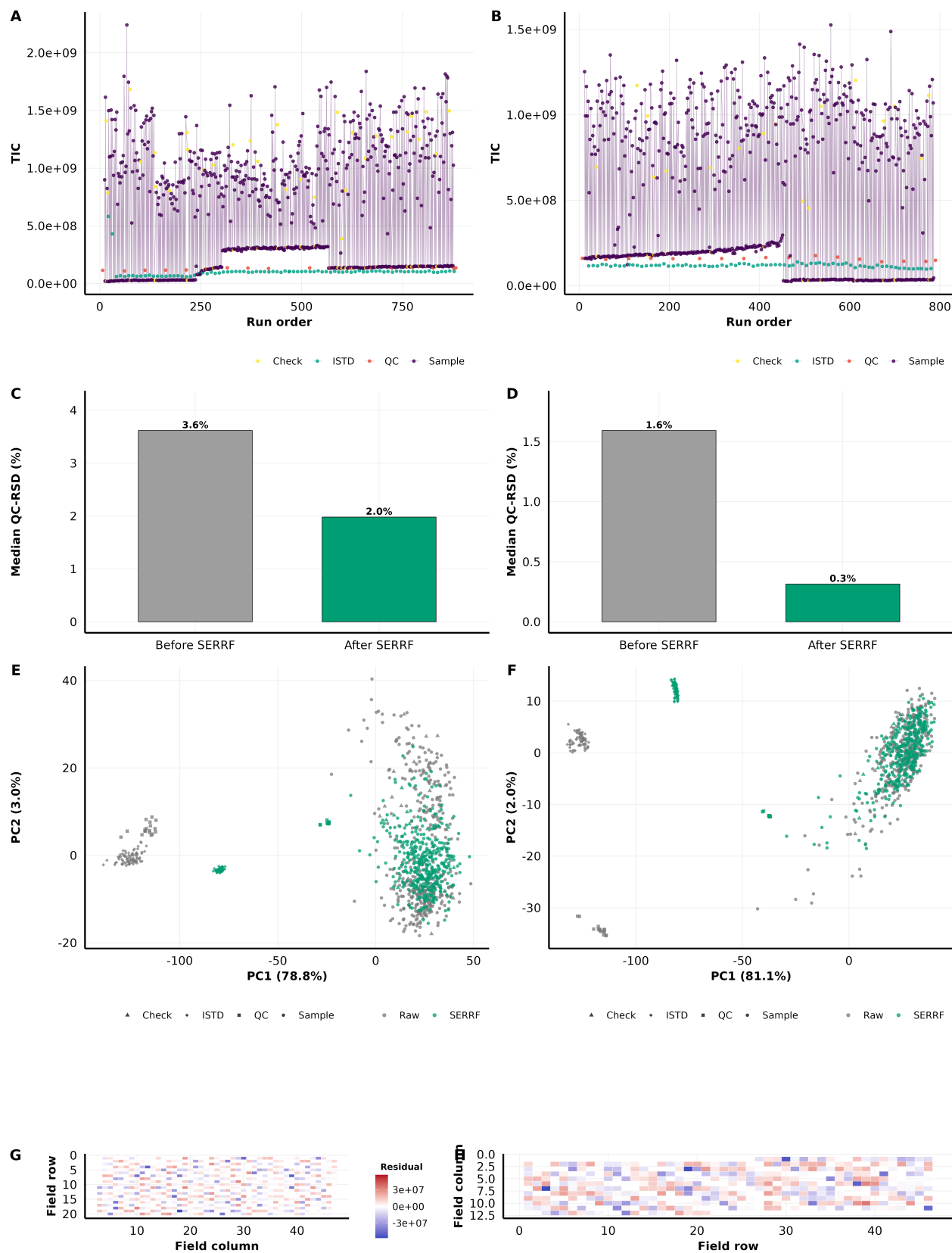
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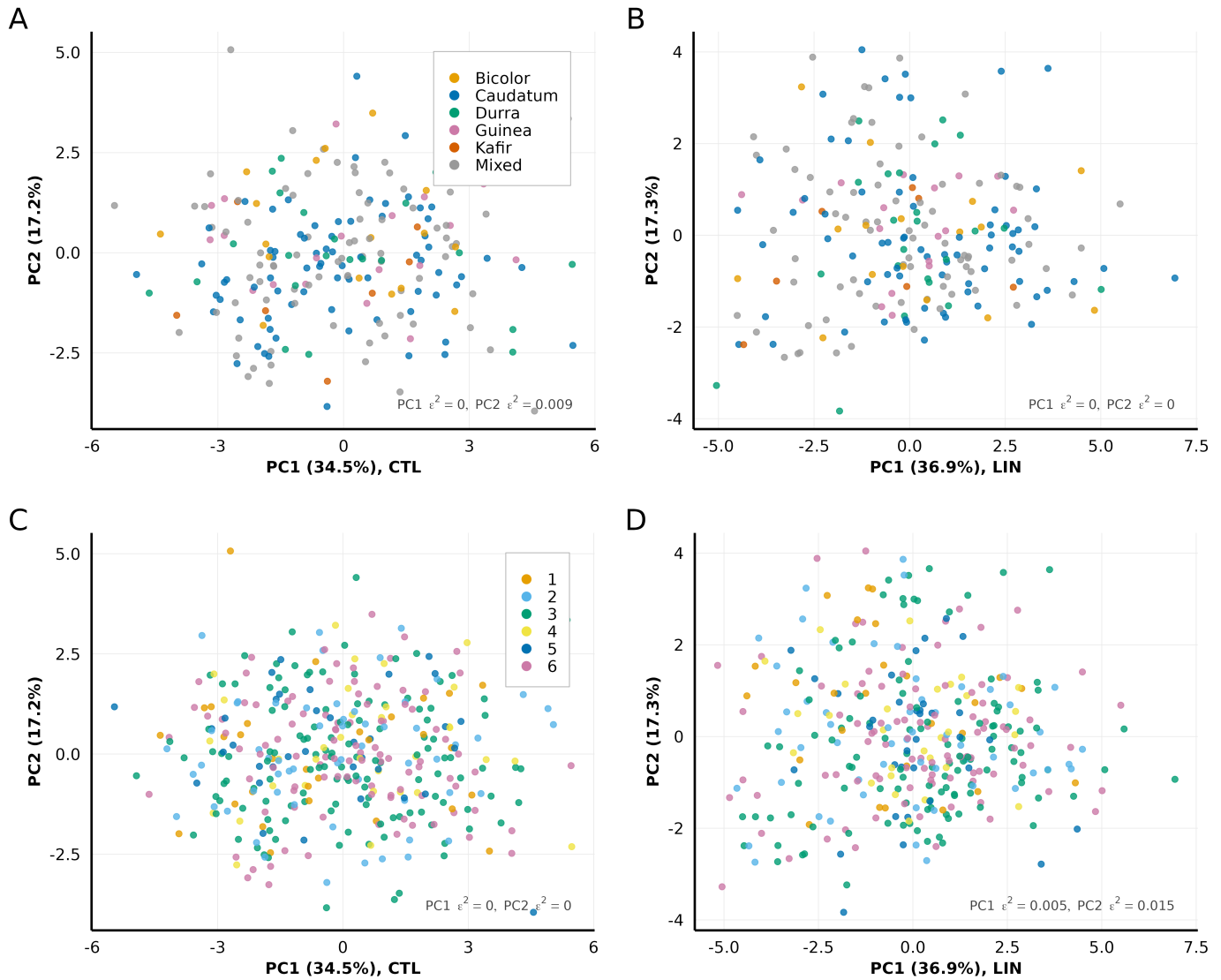
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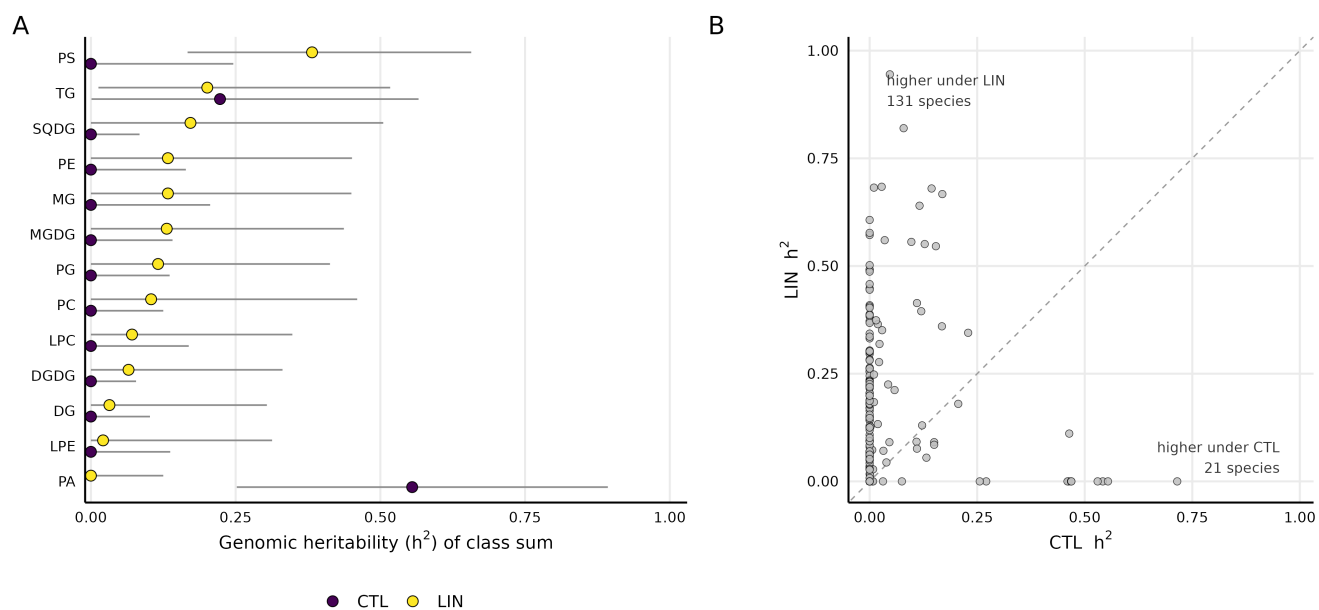
S1 Fig Field-to-database workflow. Sample collection and lipid extraction, high-resolution LC-MS lipidomic profiling, GNPS-assisted feature processing and annotation, post-acquisition correction with SERRF and SpATS, statistical description of each trial, lipidomics GWAS and candidate gene discovery, LINEX2 reaction mapping, and dissemination through the SoLD Shiny application.



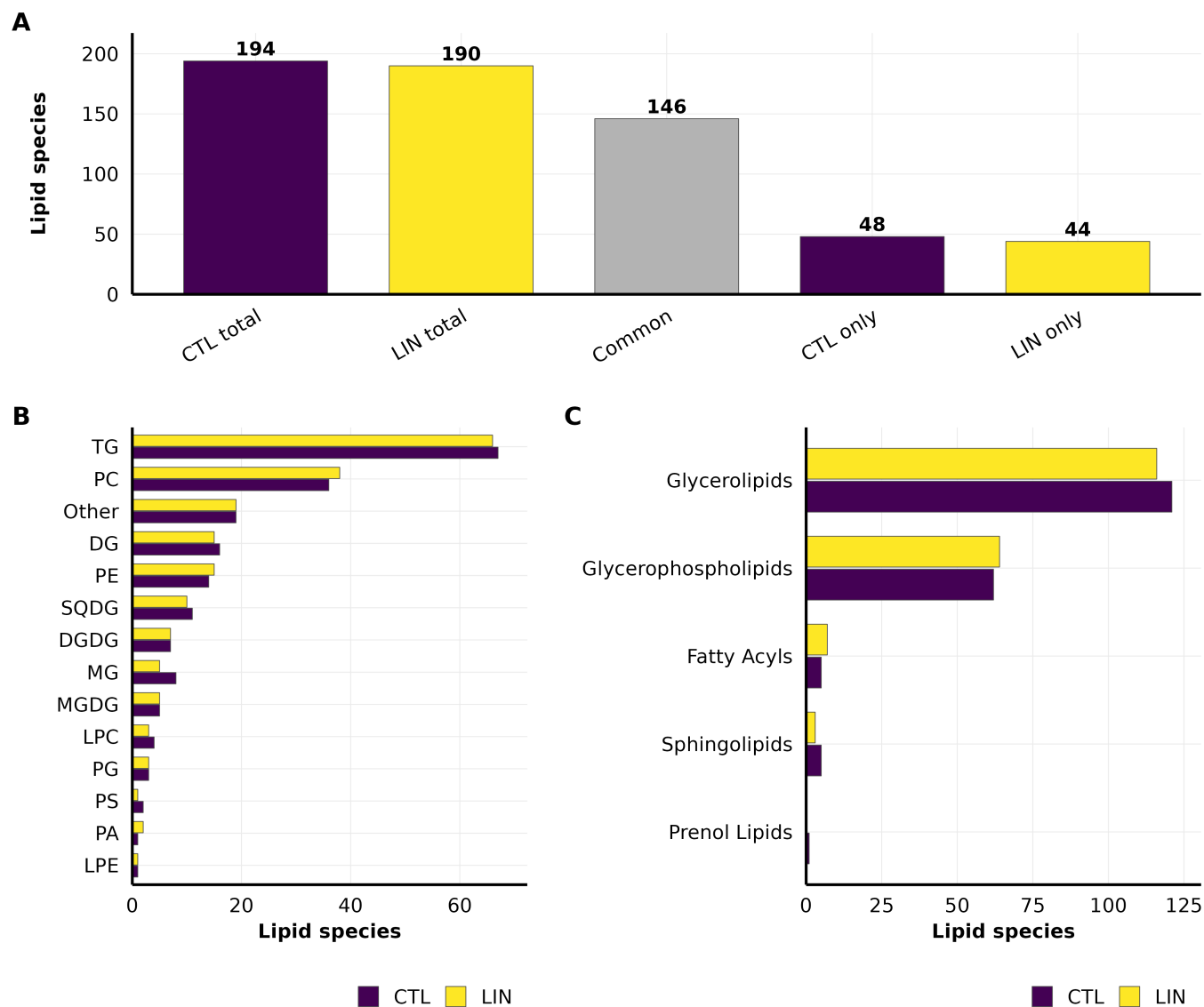
S2 Fig Quality control, SERRF normalization, and SpATS spatial adjustment diagnostics. (A,B) Run-order TIC plots for CTL and LIN conditions demonstrating stable analytical performance throughout the acquisition sequence. (C,D) QC-RSD distributions before and after SERRF normalization, showing reduction in technical variation. (E,F) PCA score plots comparing sample clustering before and after normalization. (G,H) SpATS spatial residual maps indicating successful mitigation of field-position effects in both conditions.



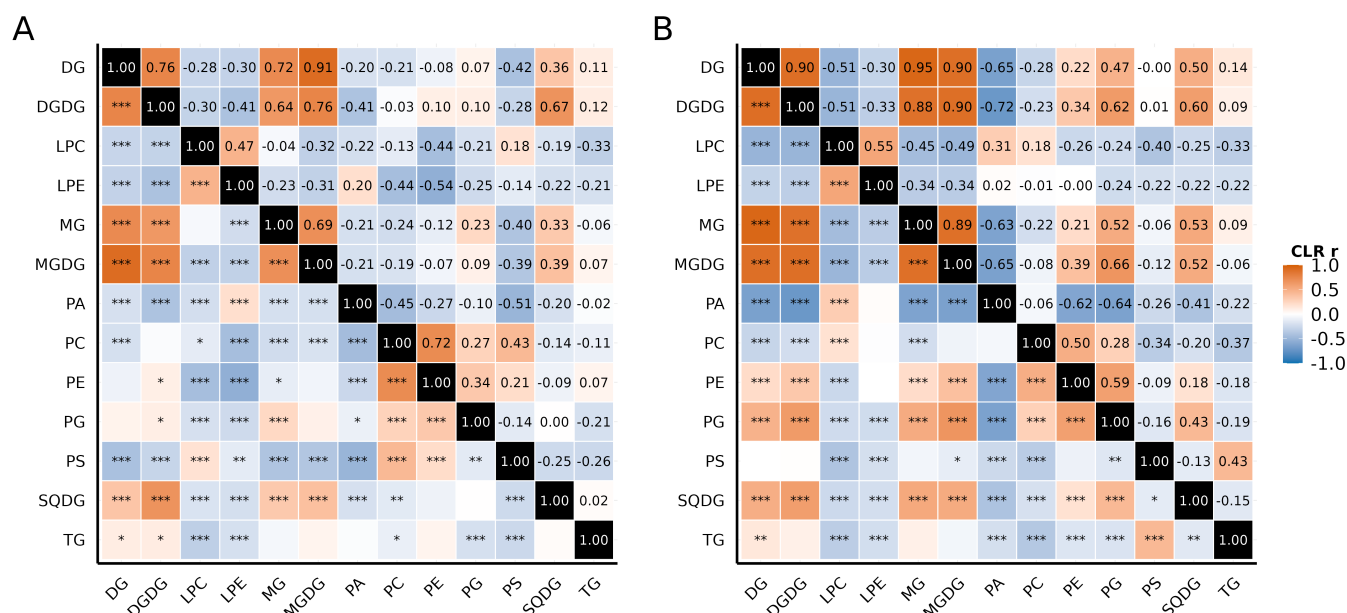
S3 Fig Lipid-class composition does not separate accessions by botanical race or by genetic cluster. Principal component analysis of the 13 lipid-class percentages, run separately within each field trial on $\log_{10}$ values scaled to unit variance so that no single dominant class sets the axes. **(A,B)** Accessions coloured by botanical race in CTL and LIN. **(C,D)** The same two ordinations coloured by marker-based genetic cluster ($K = 6$). Each panel gives the variance in PC1 and PC2 explained by that grouping (ϵ^2 , Kruskal–Wallis). No grouping explains either leading axis in either trial.



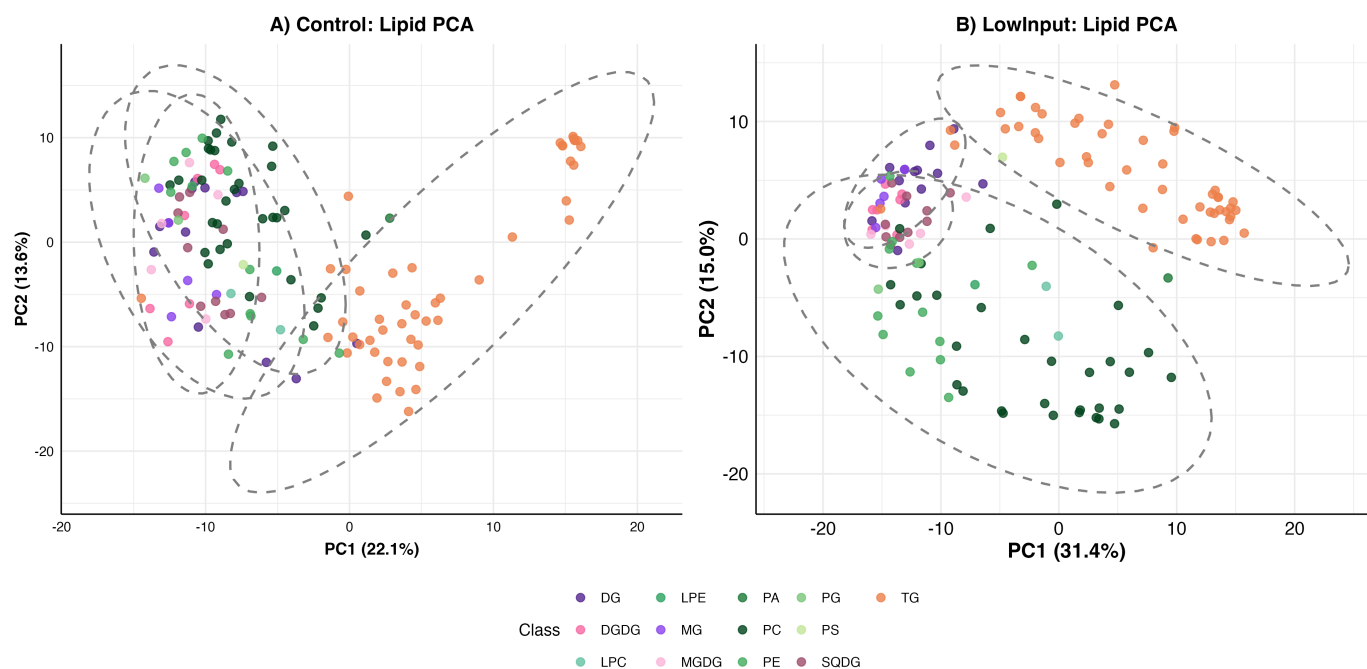
S4 Fig Genomic heritability of lipid traits in the two field trials. (A) Genomic heritability of each lipid-class sum, point estimate with its 95% profile-likelihood interval. (B) Paired per-species heritability, one point per lipid species, CTL on the horizontal axis and LIN on the vertical; the dashed line is equality. Both panels use the same genomic relationship matrix, the same 351 genotypes and the same 163 species detected in both trials, so the two trials differ only in the phenotype measured.



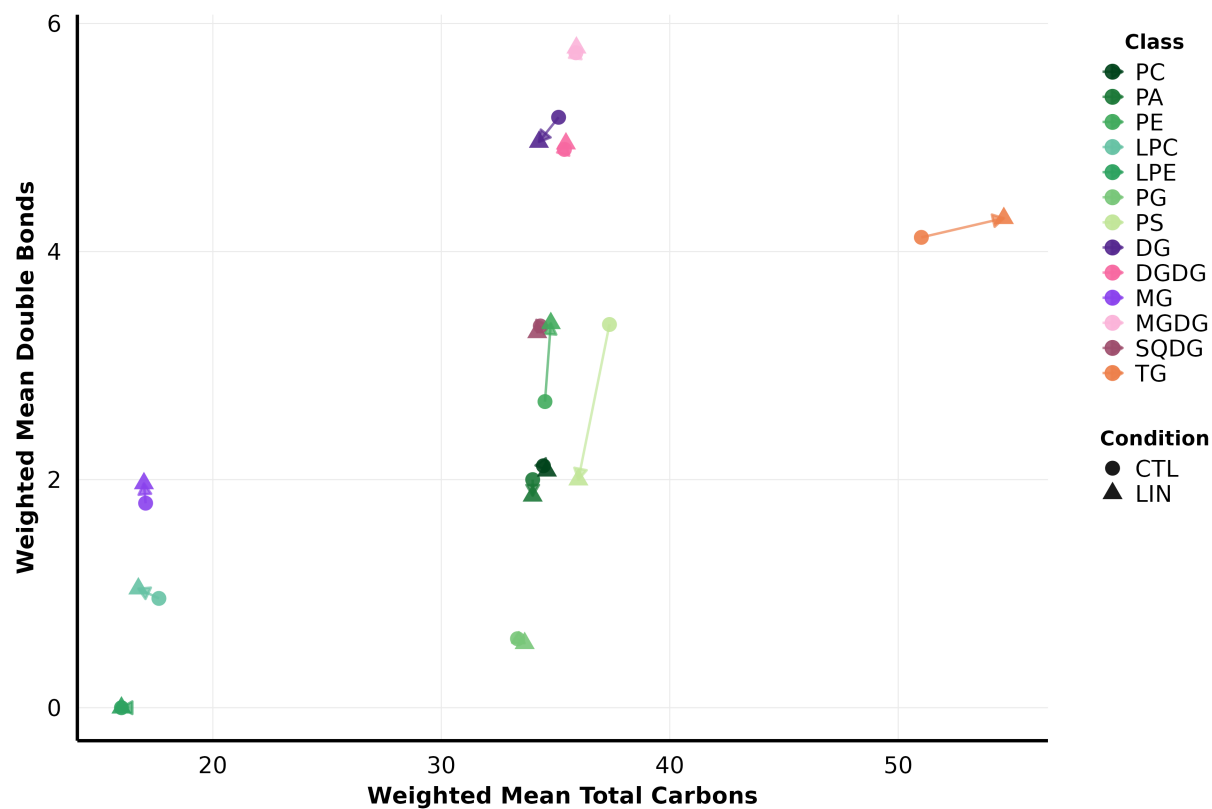
S5 Fig Lipid species counts and overlap between conditions. (A) Species detected in each trial, with the common and trial-specific counts. (B) Counts by lipid class. (C) Counts by LIPID MAPS category. Counts cover features annotated in lipid shorthand notation; the carotenoids, sterols, tocopherols, quinones and free sphingoid bases carried under trivial names are excluded here but retained in the composition analyses.



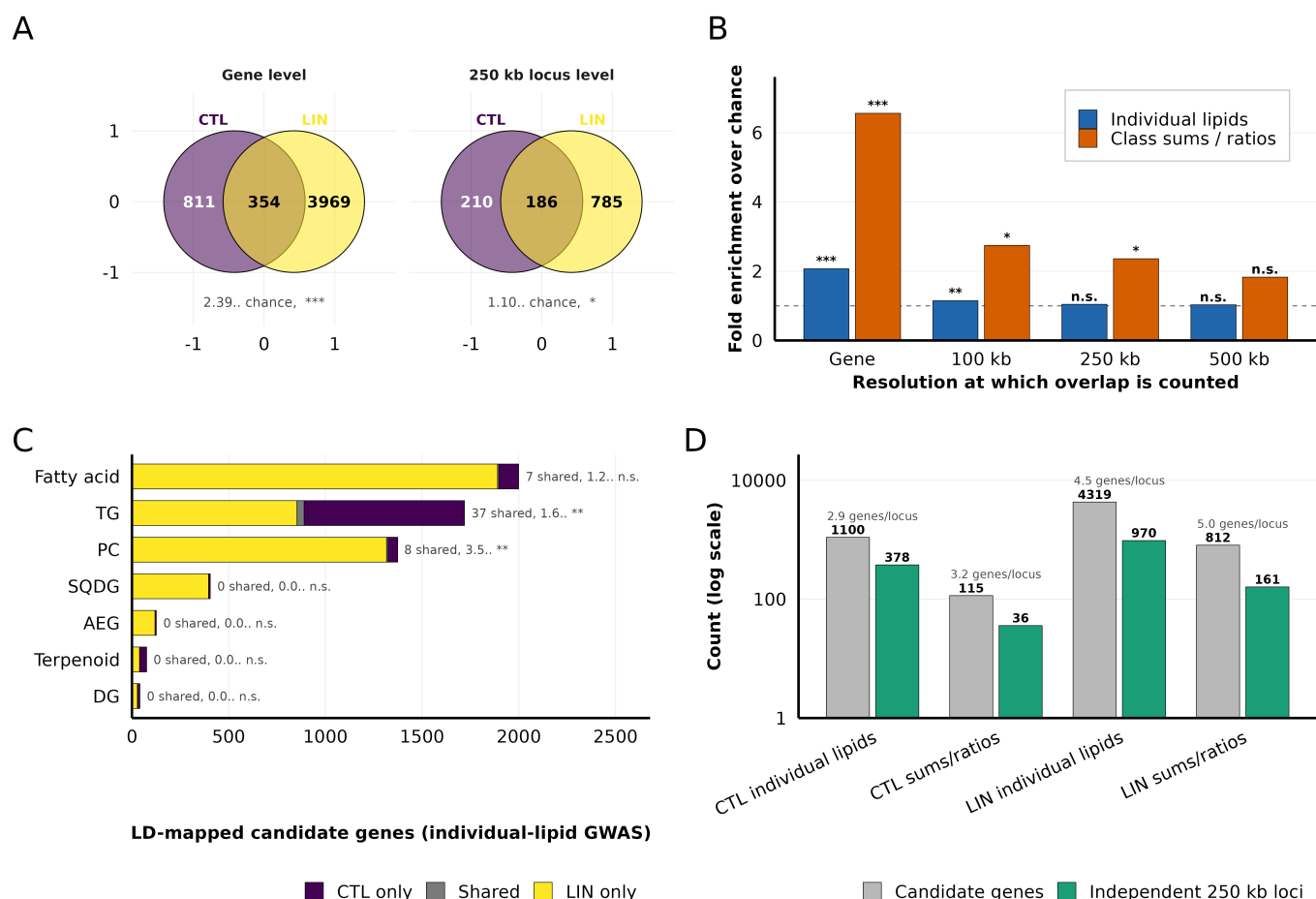
S6 Fig Correlation structure among lipid classes within each field trial. Pearson correlation matrices computed on class-level centered log-ratio abundances for (A) CTL and (B) LIN. The lower triangle gives the correlation coefficient and the upper triangle its significance (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Each matrix is computed within one trial and the two are not differenced.



S7 Fig PCA of lipid species in compositional space. (A) PCA score plot of CLR-transformed lipid species in CTL samples, with 95% confidence ellipses for LIPID MAPS category groups. (B) PCA score plot for LIN samples using the same transformation. Points are colored by lipid class.



S8 Fig Lipid classes in a reduced chemical space. Each point is one lipid class, placed at its abundance-weighted mean total carbon number and its abundance-weighted mean double-bond count across the species it contains, with the weights taken from the %TIC composition. Colour gives the class and shape the trial, and the arrow runs from a class's CTL position to its LIN position. The two trials differ in year, field and acquisition batch, so these shifts are descriptive. Classes represented by one to three species (PS, PA, LPE, PG) have unstable positions, because the species detected in the two trials are not the same; the PS shift, for example, follows the loss of PS(18:0/20:4) from the detected set rather than remodelling of a PS pool.



S9 Fig CTL and LIN GWAS identify largely non-overlapping candidate loci. (A) Overlap of candidate sets at gene level and after collapsing linkage blocks into 250 kb loci, with fold enrichment over chance and hypergeometric P . (B) Fold enrichment of the overlap as a function of the resolution at which it is counted, for individual-lipid and class-sum/ratio traits. The dashed line marks chance expectation; *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, n.s., not significant. (C) Candidate genes per lipid class in the individual-lipid GWAS, partitioned into CTL-only, shared, and LIN-only. (D) Candidate-gene counts compared with the number of independent 250 kb loci they occupy (log scale).